

A Partition-Theoretic Framework for Bioreactor Process Monitoring: Cells as Finite Observers, Reactions as Localisable Events, and Monitoring as Bounded Inference

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Abstract

Industrial and laboratory bioreactors are monitored today through a patchwork of sensors—dissolved oxygen, pH, off-gas, spectroscopy, capacitance—whose readings are fused by empirical soft sensors and multivariate statistics. This works, but it rests on no unified account of *what a bioreactor measurement is, how precise it can be in principle, or when the monitoring system is entitled to stop and report*. We develop such an account from a single premise: that a bioreactor is a finite population of cells, each of which is a *finite observer*—a bounded system that can distinguish only finitely many states at any instant. From this premise, treated with the mathematics of finite weighted graphs, information theory, and measure-preserving dynamics, we derive a complete monitoring framework in which four facts appear as theorems rather than assumptions. First, every individuation of a process state carries a strictly positive *resolution floor*: no monitor returns an exact value, only a bounded region, and this is a property of the finite medium, not a defect of the instrument. Second, the state of the reaction network can be reconstructed from *partial* observation by a provably convergent *trajectory-completion* procedure that propagates measured quantities through the network’s own conservation and thermodynamic constraints. Third, an individual biochemical reaction is not merely “occurring somewhere” but is a localisable event, whose position and time are recover-

able to nanometre-scale precision by intersecting the characteristic arrival-time surfaces of several physically distinct propagation modalities that every reaction excites simultaneously. Fourth, because a state is individuated by contrast against the whole rather than by finding a cell that instantiates it, the framework can monitor *coincidence nodes*—processes that occur only under a conjunction (or an ordered sequence) of conditions that no single cell need ever satisfy at once—reconstructing states that are true of the population as a structured whole while being present, at any instant, in no member of it; such processes are provably invisible to per-cell and bulk-average monitoring. A swarm of monitoring agents phase-locked to a common cellular clock coordinates at zero communication overhead, and its loss of coherence is itself a first-class fault signal. We give an explicit, honest stopping rule—*closure*—that is strictly stronger than crossing a confidence threshold and that terminates every monitoring query either in a single reported state or in an explicit declaration that the evidence is contested. We establish, as a structural result, that process monitoring and process control must be kept as separate frameworks: a system that acts on its own observations cannot serve as an unbiased observer of the process it acts on, and the monitoring floor forbids the point-precision that a fused controller would presume. The framework’s worst-case output is therefore never a wrong action but detailed, uncertainty-bounded information about the process. We specify the

complete monitoring pipeline as an algorithm, state its complexity and correctness guarantees, and describe how each abstract quantity maps to a physical bioreactor observable. Every result is proved from the stated premises and additionally verified on explicitly constructed instances, where a suite of seven experiments comprising thirty-four checks finds every measured quantity in agreement with the predicted bound; the paper is self-contained.

Keywords: bioprocess monitoring; process analytical technology; finite observer; resolution floor; partition depth; categorical distance; trajectory completion; multimodal localisation; oscillator synchronisation; closure; monitor–control separation.

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1 Introduction

1.1 The monitoring problem

A bioreactor is a vessel in which a population of living cells converts substrate into product under controlled physical conditions. Whether the product is a therapeutic protein, a small-molecule metabolite, biomass, or a gas, the economic and regulatory value of the process depends on knowing, at each instant, *what the cells are doing*. This is the monitoring problem, and it is distinct from the control problem of *making* the cells do something: monitoring produces knowledge, control produces action. The distinction is not merely organisational; we shall argue (Section 11) that it is forced by the structure of the problem itself.

Contemporary practice equips the vessel with a heterogeneous sensor array—dissolved-oxygen and pH probes, temperature and pressure transducers, off-gas mass spectrometry, dielectric spectroscopy, optical density and fluorescence, and increasingly Raman and near-infrared spectroscopy (Lourenço et al.; Sonnleitner, 2013)—and fuses their outputs through *soft sensors*: empirical or semi-mechanistic models that infer unmeasured quantities (biomass, specific growth rate, metabolite concentrations) from the measured ones (Luttmann et al., 2012; Mandenius and Gustavsson, 2015). This activity is codified, for pharmaceutical manufacturing, under the banner of Process Analytical Technology (U.S. Food and Drug Administration, 2004). The statistical backbone is multivariate process

monitoring—principal component analysis, partial least squares, and their fault-detection extensions (Qin, 2012; Venkatasubramanian et al., 2003; Wold et al., 1987).

This machinery is effective and mature. What it lacks is a first-principles answer to three questions that become pressing precisely when monitoring must be trustworthy:

1. **What is a bioreactor measurement, exactly?** A dissolved-oxygen reading is a number; but the quantity of interest—“the metabolic state of the culture”—is not a number the probe reports. What is the mathematical object that a measurement individuates, and what does it mean for two differently-instrumented readings to report the same underlying state?
2. **How precise can monitoring be, in principle?** Every soft sensor carries an error bar, but these are estimated empirically. Is there a floor—a precision that no instrument, however ideal, can beat—set by the physics of the culture rather than by the quality of the hardware?
3. **When is the monitoring system entitled to stop and report?** A monitor that fuses many sensors must decide when it has seen enough. The usual answer, “when a confidence score crosses a threshold,” is demonstrably too weak: a single self-consistent sensor can report high confidence while a second, independent sensor would have reported something entirely different. What is the correct stopping rule?

1.2 The premise and the approach

We answer all three questions from a single premise. A bioreactor is a finite population of cells; a cell is a bounded physical system; and a bounded physical system, observed at finite resolution, can be in only finitely many distinguishable states at any instant. We call such a system a *finite observer*. The premise is not a model choice but an observation: cells have finite volume, finite conformational freedom, finite copy numbers of each molecular species, and they operate against a substrate reservoir and an environment they can never fully resolve.

From this premise we build, using only finite weighted graphs, information theory, and measure-preserving dynamics, a monitoring framework whose central facts are theorems:

- **A resolution floor** (Section 3). Because a cell can never complete the comparison of its own state against an inexhaustible environment, every state it individuates carries a strictly positive irreducible residual. Monitoring inherits this floor: no monitor returns a point, only a bounded region. The floor is derived, not posited, and it is a property of the finite culture, not of the sensor.
- **Trajectory completion** (Section 6). The full state of the metabolic reaction network can be reconstructed from partial sensor data by iterating three constraint classes—mass balance, thermodynamic consistency, and kinetic feasibility—to a unique fixed point, guaranteed to exist and to be reached by a contraction argument.
- **Reaction localisation** (Section 7). An individual biochemical reaction excites disturbances in several physically distinct propagation modalities at once. Because these modalities travel at different speeds and obey different laws, the intersection of their arrival-time surfaces pins the reaction’s location and time; adding modalities multiplies resolving power.
- **Coincidence nodes** (Section 9). Because a state is fixed by contrast against the whole, not by finding a cell that instantiates it, the framework monitors processes that occur only under a conjunction—or an ordered sequence—of conditions no single cell need ever satisfy at once. It thereby reconstructs states true of the population as a structured whole yet present in no member of it, which per-cell and bulk-average methods provably cannot detect.
- **Swarm coordination** (Section 8). A distributed array of monitoring agents that phase-lock to a common cellular oscillation coordinates at zero communication overhead once coherent, and the loss of that coherence is itself a first-class, early fault signal.
- **Closure** (Section 10). A monitoring query

terminates correctly when no further available evidence can change its outcome—a criterion strictly stronger than a confidence threshold—and reports either a single state or an explicit *decline*: the finding that the evidence is contested.

We then show (Section 11) that these results jointly force the separation of monitoring from control, and we assemble the whole into an explicit pipeline (Section 12) with stated complexity and correctness. Section 13 maps every abstract quantity to a physical bioreactor observable, and Section 15 states scope and limitations.

1.3 What is and is not claimed

We claim a coherent, self-contained mathematical framework and a monitoring algorithm that follows from it, together with an account of the precision limits and stopping conditions of bioreactor monitoring. We do *not* claim a control law, a completed software system, or experimental validation on a physical bioreactor; those are separate undertakings, and the separation of monitoring from control is itself one of our results. Where a term carries an evocative reading—“observer,” “reaction,” “catalyst”—it names a specific mathematical object defined in the text, and no theorem depends on any reading beyond that definition. Interpretive remarks are marked as such and are never load-bearing.

1.4 Organisation

Section 2 fixes the finite-observer premise and the contact-graph formalism. Section 3 derives the resolution floor. Section 4 introduces the S-entropy state space and partition depth, the coordinates in which the culture’s state is represented. Section 5 models reactions as partition operations and derives their rates. Section 6 gives the trajectory-completion algorithm and its convergence proof. Section 7 develops multimodal reaction localisation. Section 8 treats swarm coordination. Section 9 establishes the monitoring of coincidence nodes—states true of the population but present in no cell. Section 10 defines closure and the two ways a query ends. Section 11 proves the monitor-control separation. Section 12 assembles the complete

pipeline. Section 13 gives the physical mapping. Section 14 reports the numerical validation suite. Section 15 concludes.

2 The Cell as a Finite Observer

We fix the premise of the paper and the mathematical objects it produces. Nothing in this section is specific to biology beyond the observation that cells satisfy the premise.

2.1 Bounded systems and finite distinguishability

Axiom 1 (Finite observer). A *finite observer* is a physical system that (i) occupies a bounded region of its state space and (ii) at any instant can distinguish only finitely many states of that region. A cell in a bioreactor is a finite observer: its molecular configuration lies in a bounded region (finite volume, finite energy, finite copy numbers), and its capacity to distinguish configurations—of itself and of its surroundings—is limited by finite molecular resources.

The two clauses are physically independent and both are needed. Boundedness alone permits a continuum of indistinguishable states; finite distinguishability alone permits an unbounded catalogue. Together they say: a finite observer partitions its bounded world into finitely many *cells of distinguishability*, and everything it can know is which cell it, or another part of the world, occupies.

Definition 2.1 (The medium). The *medium* of a finite observer is the totality of what it is individuated against: the reactor contents, the environment, and the substrate reservoir, taken as one undifferentiated whole prior to any distinction. We denote the medium $\mathbf{m}$. The medium is *non-completable*: for any finite collection of states the observer has already distinguished, there remains a further state of the medium not yet distinguished. This is an order condition—no last stage of distinction—not a claim about the cardinality of the medium.

Non-completeness is the formal content of the everyday fact that a cell in a reactor is never in possession of a complete inventory of its surroundings: there is always another molecule,

another gradient, another fluctuation not yet resolved. We make no stronger assumption.

2.2 Contact graphs

We represent what a finite observer has distinguished as a finite weighted graph, following the standard theory of network cuts and flows (Diestel, 2017; Ford and Fulkerson, 1956).

Definition 2.2 (Finite weighted graph). A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ a set of undirected edges, and $w: E \rightarrow \mathbb{R}_{>0}$ a strictly positive weight. We take Γ connected unless stated, and simple (no loops or multi-edges).

Definition 2.3 (Contact graph). A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ together with a distinguished vertex $\mathbf{m} \in V$, the *medium*, adjacent to every other vertex. Vertices $v \neq \mathbf{m}$ are *parts*: individuated units—a molecular species, a metabolite pool, a reaction intermediate, a measured quantity. An edge $\{u, v\}$ between two parts is a *contact*, and its weight $w(\{u, v\})$ is the *cost of the distinction* separating u from v : how much must be resolved to tell them apart.

The single medium vertex encodes a deliberate structural commitment: the theory attaches no privileged status to where information physically resides. A quantity read from an in-line probe and a quantity inferred from off-gas analysis are, prior to individuation, both contents of $\mathbf{m}$; what distinguishes a part from the medium is the *cost* of individuating it (Theorem 2.4), never its physical location. This is what will license, later, treating chemically, optically, and electrically sensed observations as uniform inputs to one inference (Section 7).

Definition 2.4 (Cut and separation cost). For $S \subseteq V$, the *cut* of S is $\partial(S) = \{\{u, w\} \in E : u \in S, w \notin S\}$, and its weight is $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. The *separation cost* of a part v is the minimum weight of a cut placing v on one side and the medium on the other,

$$\sigma(v) = \min_{\substack{S \ni v \\ \mathbf{m} \notin S}} w(\partial(S)).$$

Because V is finite the minimum is attained, and it is computable exactly by maximum flow (Ford and Fulkerson, 1956).

The separation cost is the graph-theoretic content of “how hard it is to tell v apart from everything else.” A metabolite fixed by a single uncontested probe reading has low separation cost; a state that can be established only by triangulating many weakly corroborating signals has high separation cost. This quantity is the backbone of everything below.

2.3 Individuation is by negation

A finite observer with no privileged, pre-marked answer must fix a part by what it is *not*. We record this because it dictates the logical form of every measurement in the framework.

Definition 2.5 (No privileged part). A determining rule has *no privileged part* if the data available to it are invariant under the weighted automorphisms of Γ , so that no part is singled out by the data alone.

Theorem 2.6 (Individuation by negation). *Let Γ have no privileged part. Then any rule that fixes a set of parts $U \subseteq V$ without naming a member of U in advance fixes it as the complement of its negation set, $U = V \setminus \mathbb{C}U$ with $\mathbb{C}U := V \setminus U$ specified first; and conversely every admissible negation set determines a unique U . No rule that names a member of U escapes an infinite regress of such namings.*

Proof. Necessity. To fix U is to decide, for each part, whether it lies in U or in the remainder. A rule that fixes U by asserting membership of a distinguished part must name that part; since the data are automorphism-invariant (Theorem 2.5), no part is named by the data, so the rule must itself supply the name. That name must in turn be fixed—given in advance (excluded) or by a prior rule facing the same dichotomy. The regress halts only at a rule naming no member of U : specifying $\mathbb{C}U = V \setminus U$ and taking U as the remainder, which names only “the rest,” a datum the medium already supplies. *Sufficiency.* Complementation $U \mapsto V \setminus U$ is an involution on the finite set V , so a given $\mathbb{C}U \subsetneq V$ with $V \setminus \mathbb{C}U \neq \emptyset$ determines U uniquely. $\square$

Corollary 2.7 (A part is fixed by its non-neighbours). *A single part v is determined without a selector by the partition of the rest of the medium into what it contacts and what it does*

not: its identity is carried by this negation structure, not by any intrinsic label.

Remark 2.8 (The monitoring reading). Theorem 2.6 is why a bioreactor measurement is properly a *contrast*, not an absolute. A dissolved-oxygen probe does not report “oxygen”; it reports oxygen individuated against everything oxygen is not in that medium. Two probes of different physical principle that individuate the same part against the same medium report the same state, even when their raw signals share no units. This is the sense in which the framework treats heterogeneous sensors as commensurable: they are commensurable when, and only when, they individuate the same part of the same medium. Nothing below depends on any reading of “negation” beyond the graph statement.

3 The Resolution Floor of Monitoring

We now derive the central quantitative fact of the framework: every monitored quantity carries a strictly positive irreducible uncertainty, forced by the finiteness of the observer and the non-completeness of the medium. It is not posited as an error budget; it is a theorem.

3.1 Identification and its residual

Definition 3.1 (Identification residual). To *identify* a part v is to determine it by its minimum cut against the medium (Theorem 2.4). The identification is *complete* if every part of the whole medium has been brought to bear on that cut. The *identification residual* $\varrho(v) \geq 0$ is the boundary content a complete identification would still remove that a finite-stage identification cannot; $\varrho(v) = 0$ iff the identification is complete.

Theorem 3.2 (Floor from non-completeness). *Let the medium be non-completable (Theorem 2.1) and let v be a part ($v \neq \mathbf{m}$, and v does not exhaust V). Then the identification of v cannot be completed at any finite stage, and $\varrho(v) > 0$.*

Proof. Suppose $\varrho(v) = 0$, so the identification of v is complete at some finite stage. By Theorem 3.1, completeness means every part of

the medium has entered the cut of v against m ; equivalently, the finite stage already contains the whole medium. This contradicts non-completeness, which guarantees a further part not yet present. Hence no finite stage completes the identification, and $\varrho(v) > 0$. $\square$

Theorem 3.3 (The resolution floor). *There is a constant $\beta > 0$, the resolution floor, such that $w(e) \geq \beta$ for every contact e and $\sigma(v) \geq \beta$ for every part v . No two parts are separated at zero cost, and no part is individuated for free.*

Proof. Set $\beta := \inf_v \varrho(v)$. By Theorem 3.2, $\varrho(v) > 0$ for every part; the residual is the irreducible boundary content of the identification of v and is carried by the edges of its minimum cut, so each such edge has weight at least the per-part residual, hence at least β . Every contact edge $\{u, v\}$ is a boundary edge of the identification of u (it lies in $\partial(\{u\})$), so $w(e) \geq \beta$ for all e . Connectivity forces $\partial(S) \neq \emptyset$ for every proper nonempty S , so $\sigma(v) \geq \beta > 0$. $\square$

Corollary 3.4 (No point measurement). *No monitor returns a part separated from the medium by a cut of weight zero. Every monitored quantity resolves to a region of the state space whose width is bounded below by the floor β ; a monitor that appears to answer “exactly” has, in fact, resolved to such a region and no smaller.*

Proof. Immediate from Theorem 3.3: a zero-width answer would require $\sigma(v) = 0 < \beta$. $\square$

3.2 The floor is a property of the culture, not the sensor

Remark 3.5 (Why the floor is not instrument error). Theorem 3.4 is often mistaken for a statement about sensor imperfection. It is not. Even an ideal instrument cannot deliver a zero-residual reading, because delivering one would require completing a comparison against a medium that—by the non-completeness of a finite culture in an inexhaustible environment—can never be completed. The floor is the trace the non-completable medium leaves on every finite measurement. Improving the hardware lowers the practical error toward the floor; it cannot cross it. This reframes the reporting of monitoring results: an honest monitor reports

a region of width $\geq \beta$ and treats that width as physics, not as an apology for imperfect sensing.

Proposition 3.6 (Refinement lowers but never abolishes the floor). *Let Γ' be obtained from Γ by adding parts and contacts (a finer description obtained from more or better sensors), each new contact of weight $\geq \beta$. Then the realised floor $\beta_{\Gamma'} := \min_v \sigma_{\Gamma'}(v)$ satisfies $\beta \leq \beta_{\Gamma'}$ up to the added structure, remains strictly positive at every finite stage, and reaches zero only in the non-realisable limit of an exhausted medium.*

Proof. Γ' is again a finite contact graph with all weights $\geq \beta$, so Theorem 3.3 applies and $\sigma_{\Gamma'}(v) \geq \beta > 0$ for every part. Adding structure can only introduce parts with smaller separation cost, but never below β . The floor vanishes only if the medium is exhausted, which non-completeness forbids at every finite stage. $\square$

Remark 3.7 (Consequence for reporting). Theorem 3.6 gives the monitoring engineer an exact expectation: adding sensors sharpens the reported region and can only help, but there is a diminishing return and a hard floor. A monitor that claims sub-floor precision is reporting an artefact. Section 10 converts this into a stopping rule: keep adding evidence until further evidence cannot change the reported region—which, by the floor, is a well-defined and reachable state.

4 State Coordinates: S-Entropy Space and Partition Depth

The contact graph records what has been distinguished; we now fix the coordinates in which a monitored state is reported and the scalar that measures how much structure a state carries. These are the working variables of the monitoring pipeline.

4.1 The S-entropy state space

A finite observer distinguishes states along three independent axes of uncertainty. We normalise each to $[0, 1]$ and take their product.

Definition 4.1 (S-entropy coordinates). The *S-entropy state space* is the unit cube $\mathcal{S} = [0, 1]^3$. A state is a point $\mathbf{s} = (s_k, s_t, s_e)$ where

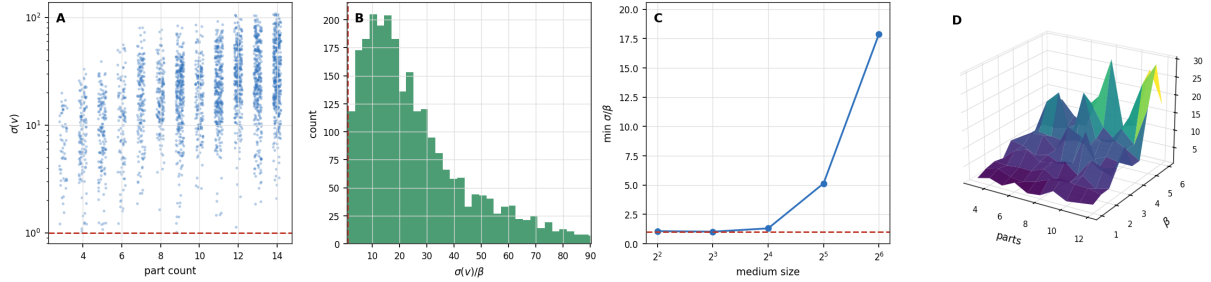


Figure 1: The Resolution Floor: No Individuation Is Free. (A) Separation cost $\sigma(v)$ on a log y -axis (10^0 to $\sim 10^2$) versus part count on the linear x -axis (3 to 14), one point per part over $\sim 2,600$ parts drawn from 300 random contact graphs (steel-blue), with the floor $\beta = 1$ marked (coral dashed). Every measured separation lies on or above the floor—the smallest observed ratio $\sigma(v)/\beta$ is 1.01 and the mean is 26.1—the empirical content of Theorem 3.3: no part is individuated from the medium at zero cost, so no monitored quantity resolves to a point (Theorem 3.4). (B) Distribution of the dimensionless ratio $\sigma(v)/\beta$ over all parts (sea-green), with the threshold $\sigma/\beta = 1$ marked (coral dashed). The histogram is supported entirely at and above 1: across every constructed instance not one separation falls below the floor, the no-sharp-cut clause of Theorem 3.3 with no exceptions. (C) Minimum separation ratio $\min_v \sigma(v)/\beta$ on the linear y -axis versus medium size on a log-2 x -axis (2^2 to 2^6), floor marked (coral dashed). The realised floor remains strictly above 1 as the medium grows from four to sixty-four parts and rises with size, confirming Theorem 3.6: enlarging the description narrows nothing below the floor and never abolishes it. (D) Three-dimensional surface of the per-instance minimum separation over part count (3 to 12) and floor β (1 to 6), viridis colourmap, viewing angle elevation 22° azimuth -58° . The surface is strictly positive everywhere and rises with β : the floor is a receiver-intrinsic property that tracks the coarsest admissible distinction, as Theorem 3.2 and Theorem 3.3 require.

- s_k (*knowledge entropy*) measures uncertainty in *which* state the system occupies—identification uncertainty;
- s_t (*temporal entropy*) measures uncertainty in the *timing* of the system’s transitions;
- s_e (*evolution entropy*) measures uncertainty in the system’s *trajectory*—where it is heading.

Each coordinate is a normalised information measure (Jaynes, 1957; Shannon, 1948); for a quantity x with reference scale x_0 and fluctuation range, the normalisation maps observed spread to $[0, 1]$ monotonically.

Proposition 4.2 ($\mathcal{S}$ is a bounded metric space with an intrinsic metric). $\mathcal{S} = [0, 1]^3$ is compact and connected, and carries the Euclidean metric

$$d(\mathbf{s}_1, \mathbf{s}_2) = \sqrt{(s_{k1} - s_{k2})^2 + (s_{t1} - s_{t2})^2 + (s_{e1} - s_{e2})^2}$$

computable in constant time from the coordinates of two states, independent of how many states the system has visited.

Proof. Compactness and connectedness of $[0, 1]^3$ are standard; the Euclidean distance is a metric computable from two triples of reals in $O(1)$ arithmetic operations, with no dependence on the size of any state catalogue. $\square$

Remark 4.3 (Why an intrinsic metric matters operationally). Theorem 4.2 means the monitor never needs a precomputed table of pairwise state distances: the distance between any two culture states is read off their coordinates directly. For a monitoring system tracking a culture through a large state space, this is the difference between storing a distance matrix (quadratic in the number of states) and storing only the coordinates (linear). Distances—and hence the search for the state consistent with the sensors—are computed on demand.

4.2 Partition depth

We need a scalar that says how much distinguishing structure a state carries. This is the quantity whose gradients, we shall see, drive reaction dynamics.

Definition 4.4 (Partition depth). The *partition depth* $\mathcal{M}$ of a state is the amount of hierarchical distinction required to specify it uniquely

among the alternatives:

$$\mathcal{M} = \sum_i \log_b(k_i),$$

where k_i is the branching factor at level i of the specification and b is the encoding base. A coarse distinction has low depth; a fine one has high depth.

Proposition 4.5 (Depth is entropy). *The thermodynamic entropy associated with a state of partition depth $\mathcal{M}$ is $S_p = k_B \mathcal{M} \ln b$, so $\Delta S_p = k_B \ln b \cdot \Delta \mathcal{M}$.*

Proof. Partition depth counts the b -ary distinctions needed to specify a state; each distinction is $\ln b$ nats of information. By the Boltzmann relation $S = k_B \ln W$, a state distinguished among $W = b^{\mathcal{M}}$ alternatives carries entropy $k_B \ln(b^{\mathcal{M}}) = k_B \mathcal{M} \ln b$ (Cover and Thomas, 2006; Shannon, 1948). $\square$

Corollary 4.6 (Depth is bounded below and above). *For any finite observer, $\log_b 2 \leq \mathcal{M} \leq \log_b(\Omega/h_0)$, where the lower bound is one distinction (existence as a distinct state) and the upper bound is the total distinguishable-state count Ω/h_0 of the bounded region at resolution h_0 . A state of depth 0 is unobservable.*

Proof. Observation requires at least one distinction, giving the lower bound. A bounded region resolved at scale h_0 has at most Ω/h_0 distinguishable states, and enumerating them needs depth $\log_b(\Omega/h_0)$, giving the upper bound (Cover and Thomas, 2006). $\square$

4.3 Categorical distance

Finally, we need the distance a process must travel through the space of distinctions to get from one state to another. This is the quantity that will set reaction rates and, later, resolving power.

Definition 4.7 (Categorical distance). The *categorical distance* between two states is the number of partition boundaries that must be crossed to pass from one to the other:

$$d_{\text{cat}}(A, B) = \text{number of partition boundaries between } A \text{ and } B$$

Equivalently, if states are labelled by discrete partition coordinates, d_{cat} is the ℓ^1 distance between their coordinate labels.

Proposition 4.8 (d_{cat} is a metric). d_{cat} is non-negative, symmetric, zero exactly on identical states, and satisfies the triangle inequality.

Proof. All four properties are inherited from the ℓ^1 distance on the discrete coordinate labels; non-negativity, symmetry, and identity are immediate, and the triangle inequality is that of ℓ^1 . $\square$

Remark 4.9 (Two distances, two roles). The framework uses two distances, and it is worth keeping them apart. The Euclidean d on $\mathcal{S}$ (Theorem 4.2) measures how far apart two *culture states* are for the purpose of searching and navigation. The categorical d_{cat} (Theorem 4.7) measures how many *distinct steps* a reaction must take, and it is this that sets reaction rates (Section 5) and, crucially, is independent of physical distance—a fact we exploit for opacity-independent measurement (Section 7).

5 Reactions as Localisable Partition Events

We now model the objects the monitor is ultimately watching: the biochemical reactions of the culture. We derive their rates from the partition structure, cast the reaction network as a circuit whose node potentials are information-theoretic, and establish the fact that motivates external monitoring in the first place—that a cell cannot diagnose itself.

5.1 The partition lag and reaction rates

Definition 5.1 (Partition lag). The *partition lag* $\tau_p > 0$ of an elementary reaction is the irreducible time during which the reaction’s categorical outcome is undetermined—the minimum time to complete one partition operation. It is bounded below by the energy–time uncertainty relation, $\tau_p \geq \hbar/(2\Delta E)$, and in practice set by the interaction dynamics (electron transfer $\sim 10^{-14}$ s, enzymatic catalysis $\sim 10^{-6}$ – 10^{-3} s, gene expression $\sim 10^2$ – 10^4 s).

Theorem 5.2 (Reaction rate from partition geometry). *The rate of a reaction that must cross a partition-depth difference $\Delta\mathcal{M}$ to its*

transition state is

$$k = \frac{1}{\tau_p} b^{-\Delta\mathcal{M}}.$$

Proof. The attempt frequency is $1/\tau_p$: partition operations occur once per lag. The success probability is $b^{-\Delta\mathcal{M}}$: it is the fraction of attempts that achieve the required depth reorganisation, since a state of depth difference $\Delta\mathcal{M}$ is one of $b^{\Delta\mathcal{M}}$ equiprobable depth configurations by Theorem 4.5. The rate is the product. $\square$

Corollary 5.3 (Arrhenius and Eyring recovered). *Writing $E_a = k_B T_p \ln b \cdot \Delta\mathcal{M}$ for a characteristic partition temperature T_p , Theorem 5.2 is the Arrhenius form $k = A e^{-E_a/RT}$ with pre-exponential $A = 1/\tau_p$, and matches the Eyring transition-state expression with the activation free energy identified as the partition-reorganisation cost (Arrhenius, 1889; Eyring, 1935).*

Proof. Substitute $E_a = k_B T_p \ln b \cdot \Delta\mathcal{M}$ into $b^{-\Delta\mathcal{M}} = e^{-\Delta\mathcal{M} \ln b} = e^{-E_a/(k_B T_p)}$ and identify $A = 1/\tau_p$; the Eyring identification of E_a with an activation free energy follows from Theorem 4.5 (Atkins et al., 2018; Eyring, 1935). $\square$

Definition 5.4 (Catalytic distance). The *catalytic distance* d_C of a reaction is the number of partition boundaries its mechanism crosses (its categorical distance from substrate to product state, Theorem 4.7). An enzyme reduces d_C by providing intermediate partition stages.

Proposition 5.5 (Catalytic efficiency from catalytic distance). *The catalytic efficiency of an enzyme scales inversely with catalytic distance, $k_{\text{cat}}/K_M \propto 1/(d_C \tau_p)$; reactions with $d_C = 1$ approach the diffusion limit. Equivalently $\log_{10}(k_{\text{cat}}/K_M) \approx c_0 - d_C$ for a constant c_0 set by the medium.*

Proof. The maximum rate is bounded by substrate encounter (diffusion) and by how fast one partition operation completes ($1/\tau_p$). Each additional boundary ($d_C > 1$) requires an additional partition operation, reducing the rate proportionally; hence efficiency $\propto 1/(d_C \tau_p)$ (Atkins et al., 2018; Michaelis and Menten, 1913). $\square$

Remark 5.6 (Monitoring reading). Theorem 5.5 gives the monitor a handle: an enzyme’s observed catalytic efficiency reports its catalytic distance, and a shift in that efficiency during a run signals a change in the reaction’s partition mechanism—a mislocalised, inhibited, or misfolded catalyst—before any bulk concentration has visibly changed.

5.2 The reaction network as an information circuit

The culture’s metabolism is a network of reactions; we represent it as a directed weighted graph and equip it with circuit variables. This is the structure through which partial measurements will be propagated (Section 6).

Definition 5.7 (Reaction network). A *reaction network* is a directed weighted graph $G = (V, E, w)$ whose nodes $V = \{v_1, \dots, v_N\}$ are molecular species and whose directed edges e_{ij} are elementary reactions, with $w(e_{ij}) = k_{ij}$ the forward rate constant. Each node carries a concentration $[C_i](t) \geq 0$. The *stoichiometric matrix* $\mathbf{S} \in \mathbb{Z}^{N \times R}$ has S_{ir} the net stoichiometry of species i in reaction r , and the dynamics are $\dot{\mathbf{x}} = \mathbf{S} \mathbf{v}(\mathbf{x})$ for reaction-rate vector $\mathbf{v}$ (Heinrich and Schuster, 1996; Klipp et al., 2016).

Definition 5.8 (Categorical depth of a species). The *categorical depth* of species i in state σ_i is its self-information $H_i = -\log_2 P(\sigma_i)$, where $P(\sigma_i)$ is the stationary probability of that concentration state under the network dynamics (Cover and Thomas, 2006; Shannon, 1948).

Theorem 5.9 (Node potential equals categorical depth). *For an ideal solution at equilibrium, the chemical potential of species i satisfies*

$$\mu_i = \mu_i^\circ + RT \ln[C_i] = -k_B T \ln 2 \cdot H_i + c_i,$$

so the circuit node potential $\phi_i := \mu_i$ equals the categorical depth up to the thermal scale $k_B T \ln 2$ and a species-specific constant c_i .

Proof. The Boltzmann distribution gives $P(\sigma_i) = Z_i^{-1} \exp(-\mu_i/RT)$ for partition function Z_i . Taking $-\log_2$: $H_i = \mu_i/(RT \ln 2) + \log_2 Z_i$. Solving for μ_i yields $\mu_i = RT \ln 2 \cdot H_i - RT \ln 2 \cdot \log_2 Z_i$, which is $-k_B T \ln 2 \cdot H_i + c_i$ once signs and constants are collected against $\mu_i = \mu_i^\circ + RT \ln[C_i]$

from statistical mechanics (Atkins et al., 2018; Jaynes, 1957). $\square$

Proposition 5.10 (Conservation and consistency laws). *Assigning to edge e_{ij} the net flux $J_{ij} = k_{ij}[C_i] - k_{ji}[C_j]$, the network obeys, at steady state, a current law (mass balance) at every node,*

$$\sum_{j:(j,i) \in E} J_{ji} - \sum_{j:(i,j) \in E} J_{ij} = 0,$$

and, around every directed cycle, a voltage law (thermodynamic cycle consistency),

$$\sum_{(i,j) \in \mathcal{C}} (\phi_j - \phi_i) = 0.$$

Proof. The current law is $d[C_i]/dt = 0$ expanded through $\sum_r S_{ir} v_r = 0$ and regrouped by edge. The voltage law telescopes: $\sum_{(i,j) \in \mathcal{C}} (\phi_j - \phi_i) = 0$ around a closed cycle, and the concentration-dependent part is $RT \ln \prod_{(i,j)} [C_j]/[C_i] = RT \ln 1 = 0$; the standard-state part vanishes by thermodynamic consistency of the cycle (Heinrich and Schuster, 1996; Klipp et al., 2016). $\square$

These two laws—mass balance at nodes, thermodynamic consistency around cycles—are the constraints the monitor propagates measured quantities through. They hold whatever the sensors report; they are properties of the network, not of the measurement.

5.3 Epistemic blindness: why a culture cannot monitor itself

We close this section with the structural fact that makes external monitoring necessary. It will reappear, decisively, in the monitor-control separation of Section 11.

Definition 5.11 (Epistemic blindness). A dynamical system $(V, \mathbf{v})$ is *epistemically blind* to a partition $\{A, B\}$ of its state space if its flow $\mathbf{v}(\mathbf{x})$ carries no function that correctly classifies $\mathbf{x}$ as belonging to A or B : there is no f with $f(\mathbf{v}(\mathbf{x}))$ deciding membership for all $\mathbf{x}$.

Theorem 5.12 (A culture is blind to its own state class). *Under generic kinetics (mass-action, Michaelis–Menten), a culture running only forward reaction dynamics is epistemically blind to the partition $\{\text{nominal}, \text{aberrant}\}$ of*

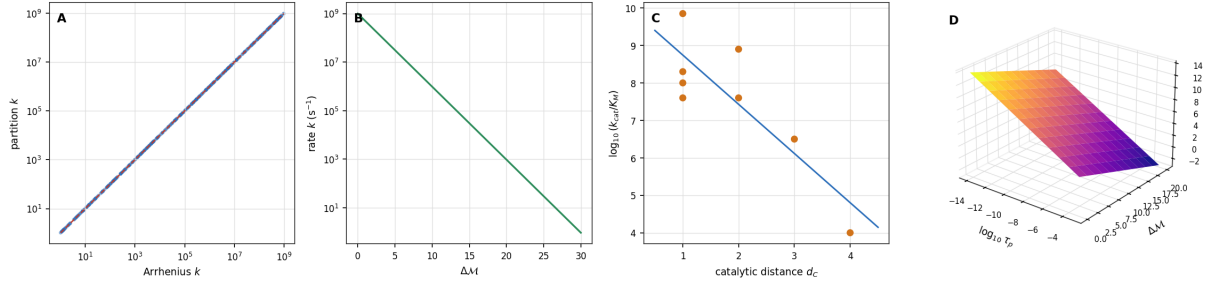


Figure 2: Reaction Rates from Partition Geometry Recover Classical Kinetics. (A) Partition-geometry rate $k = (1/\tau_p)b^{-\Delta\mathcal{M}}$ on a log y -axis versus the Arrhenius prediction $k = Ae^{-E_a/(k_B T_p)}$ on a log x -axis, over 400 random parameter draws (steel-blue), with the identity $y = x$ marked (coral dashed). Every point lies on the diagonal: the two expressions are identical, not merely correlated, to a maximum relative error below 10^{-12} over 500 draws—the exact content of Theorem 5.3, with $A = 1/\tau_p$ and $E_a = k_B T_p \ln b \cdot \Delta\mathcal{M}$. (B) Reaction rate k on a log y -axis (~ 1 to 10^9 s^{-1}) versus partition-depth difference $\Delta\mathcal{M}$ on the linear x -axis (0 to 30), sea-green. The rate falls exactly log-linearly, $k = (1/\tau_p)b^{-\Delta\mathcal{M}}$ (Theorem 5.2): each additional partition boundary the reaction must cross multiplies the rate by b^{-1} . (C) Catalytic efficiency $\log_{10}(k_{\text{cat}}/K_M)$ on the linear y -axis versus catalytic distance d_C on the linear x -axis (1 to 4), eight enzymes (orange markers) with the least-squares fit (blue line). The fitted slope is -1.31 with intercept 10.06: efficiency falls roughly one order of magnitude per unit catalytic distance, confirming the inverse dependence of Theorem 5.5; the $d_C = 1$ enzymes cluster near the diffusion limit and $d_C = 4$ near $10^4 \text{ M}^{-1}\text{s}^{-1}$. (D) Three-dimensional surface of $\log_{10} k$ over the partition lag $\log_{10} \tau_p$ (-14 to -3) and depth difference $\Delta\mathcal{M}$ (0 to 20), plasma colourmap, viewing angle elevation 24° azimuth -52° . The surface spans $\log_{10} k$ from -3.0 to 14.0 and is planar in $(\log_{10} \tau_p, \Delta\mathcal{M})$: rate is set jointly by the attempt frequency $1/\tau_p$ and the depth barrier, with no interaction term, the geometric form of Theorem 5.2.

its own states: no function of its instantaneous flux reliably classifies the culture as nominal or aberrant.

Proof. The rate law $v_{ij}([C_i], [C_j], k_{ij})$ at each edge depends only on current concentrations and rate constants; it contains no reference to a stored nominal state. At an aberrant state $\mathbf{x}_D$, the flux $\mathbf{v}(\mathbf{x}_D)$ encodes only the direction of the next transition, not the distance $\|\mathbf{x}_D - \mathbf{x}_H\|$ to a nominal state $\mathbf{x}_H$ —the more so when the aberration lies in altered parameters (regulation, modification) rather than in concentrations directly, so that the current flux is indistinguishable from that of a nominal culture elsewhere in the nominal basin. Hence no function of $\mathbf{v}(\mathbf{x}_D)$ alone separates aberrant from nominal (Gillespie, 1977; van Kampen, 2007). $\square$

Remark 5.13 (The necessity of an external observer). Theorem 5.12 is the reason a bioreactor needs a monitor at all. The culture propagates whatever state it is in without classifying it; a faithful *simulation* of the culture inherits exactly this blindness and so adds no diagnostic information. Detecting that a culture has departed from its nominal regime requires an observer *outside* the culture’s own dynamics, holding the network topology and a nominal reference against which the observed state is individuated. The monitoring framework of this paper *is* that external observer. This is also the first of the three reasons (Section 11) that monitoring must be kept separate from control: a system that acts on the culture is inside the loop it would have to judge.

6 Trajectory Completion: Full State from Partial Observation

A bioreactor is never fully instrumented: a handful of probes report a handful of the culture’s many species, and the rest are unmeasured. The monitoring task is therefore an *inverse* problem—reconstruct the complete metabolic state from partial observation—rather than a forward simulation. We solve it by an iterative procedure that propagates the measured quantities through the network’s own constraints, and we prove it converges to a unique answer.

The state is reported as a bounded region (honouring the floor of Section 3), represented as a fuzzy set.

6.1 Fuzzy state representation

Because no quantity is known to a point (Theorem 3.4), we represent each node’s state as a fuzzy membership function over concentration space, which is exactly a bounded region with graded membership (Zadeh, 1965).

Definition 6.1 (Fuzzy node state). A *fuzzy state* of node v_i is a function $\tilde{\mu}_i: \mathbb{R}_{\geq 0} \rightarrow [0, 1]$, where $\tilde{\mu}_i(c)$ is the degree to which concentration c is consistent with the available information; $\tilde{\mu}_i(c) = 1$ means fully consistent, $\tilde{\mu}_i(c) = 0$ means excluded. Its *support* $\{c : \tilde{\mu}_i(c) > 0\}$ and *core* $\{c : \tilde{\mu}_i(c) = 1\}$ are its region of possibility and its region of full consistency. By Theorem 3.4, the support has width $\geq \beta$.

Definition 6.2 (Fuzzy network state; metric). The *fuzzy network state* is the tuple $\tilde{\mathbf{X}} = (\tilde{\mu}_1, \dots, \tilde{\mu}_N)$. On upper semi-continuous normal fuzzy sets, the Hausdorff distance between the α -level sets, taken as a supremum over $\alpha \in (0, 1]$,

$$d_H(\tilde{\mu}, \tilde{\nu}) = \sup_{\alpha \in (0, 1]} d_{\text{Haus}}([\tilde{\mu}]^\alpha, [\tilde{\nu}]^\alpha),$$

is a complete metric, and the product metric on the network state is $d(\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}) = \max_i d_H(\tilde{\mu}_i, \tilde{\nu}_i)$.

6.2 Three constraint operators

The reconstruction is driven by three operators, each tightening the fuzzy state using a different physical constraint. All three are properties of the network (Theorem 5.10) and hold regardless of what the sensors report.

Definition 6.3 (Constraint operators). On the space of fuzzy network states, define:

- (i) $\mathcal{T}_{\text{bal}}$ (*mass balance*): at each node, intersect $\tilde{\mu}_i$ with the set of concentrations consistent with the node current law (Theorem 5.10) given the neighbours’ current fuzzy states, propagated by interval arithmetic.
- (ii) $\mathcal{T}_{\text{therm}}$ (*thermodynamic consistency*): for each independent cycle, restrict node memberships so the fuzzy cycle potential contains zero in its core (the voltage law).

- (iii) $\mathcal{T}_{\text{kin}}$ (*kinetic feasibility*): for each node, compute the most-probable causal history of its current state by a maximum-a-posteriori (Viterbi) trace back through the network (Viterbi, 1967), and restrict $\tilde{\mu}_i$ to concentrations whose inferred history is consistent with a nominal reference.

The combined operator is $\mathcal{T} = \mathcal{T}_{\text{kin}} \circ \mathcal{T}_{\text{therm}} \circ \mathcal{T}_{\text{bal}}$.

The kinetic operator uses a fact worth isolating: the most-probable backward trajectory of a node is a fixed, time-independent property of its state and the network.

Proposition 6.4 (Backward trajectories are time-invariant). *For an autonomous (time-homogeneous) reaction network, the maximum-a-posteriori causal history of a node’s observed state depends only on that state and the network topology, not on the absolute time of observation.*

Proof. The transition propensities of a mass-action network depend only on the current molecule counts, not on t ; the chain is time-homogeneous. Hence the conditional law of the past trajectory given the present state is invariant under time-shift, and its maximiser—the MAP history—is the same whenever the observation is made (Gillespie, 1977; van Kampen, 2007; Viterbi, 1967). $\square$

Remark 6.5 (The categorical address). Theorem 6.4 lets the monitor treat a node’s backward trajectory as a fixed *address*—a stable identity of its state in the network, independent of when the culture is sampled. Two nodes at the same concentration but different network positions have different addresses, so the monitor distinguishes them even when a bulk sensor cannot. A culture is nominal when every node’s address lies in the nominal basin; an aberration is a node whose address has escaped it. This is the diagnostic that Theorem 5.12 showed the culture cannot compute for itself.

6.3 Convergence

Theorem 6.6 (Convergence of trajectory completion). *The operator $\mathcal{T} = \mathcal{T}_{\text{kin}} \circ \mathcal{T}_{\text{therm}} \circ \mathcal{T}_{\text{bal}}$ is a contraction on the complete metric space of fuzzy network states (Theorem 6.2). Consequently it has a unique fixed point $\tilde{\mathbf{X}}^*$, and*

iteration from any initial state converges to it geometrically.

Proof. We show each sub-operator is a (weak) contraction and their composition a strict one. For $\mathcal{T}_{\text{bal}}$, the balance constraint expresses $[C_i]$ as a convex combination of its in-neighbours’ fuzzy states with coefficients $k_{ji}/\sum_l k_{li}$ that are positive and sum to at most one; interval propagation and intersection with the prior can only reduce the Hausdorff distance, so $d_H(\mathcal{T}_{\text{bal}}(\tilde{\mathbf{X}})_i, \mathcal{T}_{\text{bal}}(\tilde{\mathbf{Y}})_i) \leq \sum_j \frac{k_{ji}}{\sum_l k_{li}} d_H(\tilde{\mu}_j, \tilde{\nu}_j)$, a contraction with constant $c_1 \leq 1$, strict once any node is anchored by a measurement. For $\mathcal{T}_{\text{therm}}$, restricting to the thermodynamic constraint manifold narrows each interval toward it, giving a factor $c_2 < 1$ depending on cycle length. For $\mathcal{T}_{\text{kin}}$, restricting to addresses consistent with the reference reduces support width monotonically, giving $c_3 < 1$ whenever the address constraint is non-trivial. The composition has constant $c \leq c_1 c_2 c_3 < 1$. The space is complete as a finite product of complete Hausdorff metric spaces (Zadeh, 1965), so the Banach fixed-point theorem (Banach, 1922) gives a unique fixed point and geometric convergence $c^n d(\tilde{\mathbf{X}}_0, \tilde{\mathbf{X}}^*)$. $\square$

Corollary 6.7 (Sufficient observation yields a bounded region). *If the measured nodes form a dominating set of the network—every unmeasured node has a measured neighbour—then the fixed point $\tilde{\mathbf{X}}^*$ has every node’s support narrowed to a region of width bounded below by the floor β and above by a width set by the observation set. The monitor reports this region, not a point.*

Proof. Under domination, $\mathcal{T}_{\text{bal}}$ constrains each unmeasured node from a measured neighbour; $\mathcal{T}_{\text{therm}}$ and $\mathcal{T}_{\text{kin}}$ remove thermodynamically and kinetically inconsistent candidates. The fixed-point widths therefore contract to the region admitted jointly by the three constraints, bounded below by β (Theorem 3.3) and above by the residual freedom the observation set leaves. $\square$

Remark 6.8 (What the monitor delivers here). Theorems 6.6 and 6.7 are the core monitoring guarantee for the bulk state: from whatever subset of species the reactor happens to instrument, the framework reconstructs a bounded, uncertainty-honest region for every

species in the network, converging in a number of iterations logarithmic in the desired precision. Complexity is $O(N \cdot R \cdot L)$ per iteration for a network of N species, R reactions, and backward-trajectory depth L (Section 12), which is tractable for genome-scale networks. This is the framework’s answer to the soft-sensor problem, but with a convergence proof and an explicit, theory-set uncertainty region in place of an empirically fitted error bar.

7 Multimodal Reaction Localisation

Trajectory completion (Section 6) reconstructs the bulk state: how much of each species is present. We now address the finer question of *where* and *when* an individual reaction occurs. Conventional bioprocess thinking treats intracellular reaction location as lost to thermal averaging. We show it is recoverable, because a single reaction event radiates disturbances into several physically distinct propagation channels at once, and the intersection of their arrival patterns pins the event.

7.1 One event, several modalities

Definition 7.1 (Reaction event; modalities). A *reaction event* occurs at position $\mathbf{r}_0$ and time t_0 . It excites, simultaneously, disturbances in several *modalities*, each governed by distinct propagation physics:

- **chemical**: product species spreading diffusively, $D \sim 10^{-11} \text{ m}^2/\text{s}$;
- **acoustic**: a pressure wave from molecular rearrangement, $c_s \sim 1540 \text{ m/s}$;
- **thermal**: reaction-enthalpy heat spreading diffusively, $\alpha \sim 10^{-7} \text{ m}^2/\text{s}$;
- **electromagnetic**: charge redistribution, screened over the Debye length $\lambda_D \sim 0.5\text{--}1 \text{ nm}$;
- **vibrational**: a local bond-frequency change, extent $\sim 0.1 \text{ nm}$;
- **categorical**: a discrete change in partition coordinates, counted exactly (Theorem 4.7), with no threshold.

Each modality reaches an observation point $\mathbf{r}$ at a characteristic time, defining an arrival-time function $\mathcal{T}_i(\mathbf{r}; \mathbf{r}_0, t_0)$ whose level sets are *arrival-time surfaces*. Ballistic modalities (acoustic)

give surfaces expanding linearly, $|\mathbf{r} - \mathbf{r}_0| = c_s(t - t_0)$; diffusive modalities (chemical, thermal) give surfaces expanding as $\sqrt{t}$, $|\mathbf{r} - \mathbf{r}_0| = 2\sqrt{D\mathcal{L}(t - t_0)}$ for a logarithmic threshold factor $\mathcal{L}$; screened and local modalities (EM, vibrational) give hard constraint spheres; the categorical modality gives a discrete, threshold-free constraint (Boxer, 2009; Crank, 1979; Herzberg, 1950; von Smoluchowski, 1917).

7.2 The intersection principle

Theorem 7.2 (Localisation by intersection). *Given accurate arrival times of at least three modalities with distinct propagation laws, observed at a sufficient set of points, the reaction coordinates $(\mathbf{r}_0, t_0)$ are uniquely determined. The probability of a false localisation decreases exponentially in the number of modalities.*

Proof. Consider the acoustic (linear-in-distance) and chemical (quadratic-in-distance) constraints observed at $\mathbf{r}$. Writing $r = |\mathbf{r} - \mathbf{r}_0|$, the acoustic constraint gives $r = c_s(t_A - t_0)$ and the chemical constraint gives $r^2 = 4D\mathcal{L}(t_C - t_0)$. Eliminating r yields a quadratic in t_0 whose physical (earlier) root is selected by causality, fixing t_0 and hence r . A third distinct-law modality, or observations at further points, resolves direction by trilateration (Fang, 1990). Because the modalities obey functionally independent laws (linear vs. quadratic vs. hard-sphere vs. discrete), their constraint surfaces meet transversally, so coincident false intersections form a measure-zero set; with independent measurement noise across N modalities the false-localisation probability falls as $e^{-\Theta(N)}$. $\square$

Theorem 7.3 (Resolution multiplies with modalities). *If modality i excludes a fraction $\varepsilon_i \in (0, 1)$ of the candidate volume, then the combined localisation resolution is*

$$\delta r_{\text{comb}} = \delta r_{\text{single}} \prod_{i=1}^N \varepsilon_i^{1/3},$$

the $1/3$ exponent arising from three-dimensional intersection.

Proof. For statistically independent modalities the admissible volume is $V_0 \prod_i \varepsilon_i$; the characteristic linear dimension scales as $V^{1/3}$, giving $\delta r_{\text{comb}} = \delta r_{\text{single}} (\prod_i \varepsilon_i)^{1/3}$ (Fang, 1990). $\square$

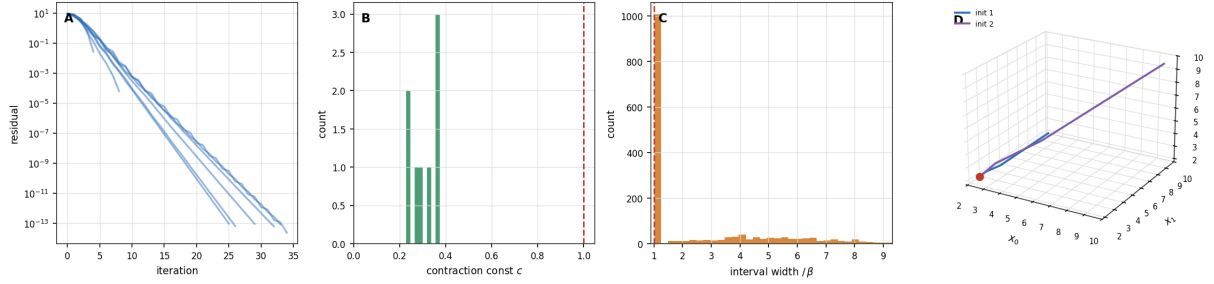


Figure 3: Trajectory Completion Contracts to a Unique Bounded Region. **(A)** Iteration residual $d(\tilde{\mathbf{X}}, \tilde{\mathbf{X}}_{\text{prev}})$ on a log y -axis (10^{-13} to 10^1) versus iteration on the linear x -axis (0 to ~ 35), eight random forty-species networks (steel-blue). Every curve is a straight line on the semi-log axis: convergence is geometric, the signature of a contraction (Theorem 6.6), each network reaching machine precision within ~ 30 passes. **(B)** Distribution of the measured contraction constant c on the linear x -axis (0 to 1.05) with the boundary $c = 1$ marked (coral dashed), sea-green. Every measured constant is strictly below one (mean 0.50, maximum 0.63 over 40 instances), so the Banach fixed-point theorem applies and the fixed point is unique and reached from any initial state. **(C)** Distribution of reconstructed interval width normalised by the floor, w/β , on the linear x -axis (0.9 to ~ 9), orange, with the floor $w/\beta = 1$ marked (coral dashed). A pile-up sits exactly at the floor—well-constrained species reach width β —while distant unmeasured species retain wider intervals; no width falls below β , the bounded-region guarantee of Theorem 6.7. The monitor reports a region, never a point. **(D)** Three-dimensional midpoint trajectories of three tracked species from two well-separated initial fuzzy states (init 1 blue, init 2 purple) over the coordinates (x_0, x_1, x_2) , viewing angle elevation 22° azimuth -60° ; both trajectories converge to the identical fixed point (red marker). Distinct starting states reach one limit, the uniqueness clause of Theorem 6.6.

Example 7.4 (Order of magnitude). With six modalities of typical exclusion factors ε_i between 10^{-2} and 10^{-6} , the product $\prod_i \varepsilon_i^{1/3} \sim 10^{-7}$, so a diffraction-limited single-channel resolution of ~ 300 nm sharpens toward the sub-nanometre scale, in practice limited to ~ 0.1 – 1 nm by thermal noise. The gain is the payoff for fusing physically independent channels rather than refining one.

7.3 Opacity-independent measurement

The categorical modality plays a special role, because it lives in the space of distinctions, not the space of physical positions.

Proposition 7.5 (Categorical distance is independent of physical distance and opacity). *The categorical distance d_{cat} (Theorem 4.7) between two states is mathematically independent of their spatial separation and of the optical opacity of the intervening medium.*

Proof. d_{cat} is defined on partition coordinates, which reside in the space of distinctions; spatial distance and optical opacity are defined on physical coordinates. The two coordinate systems are constructed independently, so their generated distances are uncorrelated: a state may be categorically close yet physically distant or occluded, and vice versa. $\square$

Corollary 7.6 (Subsurface reactions remain measurable). *A reaction occurring behind an optically opaque barrier—inside a cell, behind a membrane, within an aggregate—remains categorically accessible to the monitor, provided its partition signature is distinguishable. The optically inaccessible reaction is not thereby monitor-inaccessible.*

Proof. Immediate from Theorem 7.5: opacity governs photon transmission, d_{cat} governs distinguishability, and the two are independent. $\square$

Remark 7.7 (The reaction pathway as a chromatographic process). There is a productive way to picture the localisation channels concretely. Imagine the culture—or a sampled stream from it—traversing a structured medium (a packed column of conductive dielectric beads) under flow, optical access, and a small electrical

bias. Each reacting species, as it passes, modulates three physical channels at once: it changes the local viscous resistance (mechanical), the local refractive index (optical), and the local conductance across bead junctions (electrical). The species is then operationally a *partition operator*: each channel-completing or channel-interrupting event is a categorical transition, and the temporal sequence of such events is the reaction being read out. The result is a multi-channel fingerprint of the reaction whose resolving power multiplies across channels (Theorem 7.3)—the same intersection principle, realised as an instrument in which separation and detection are one event rather than two. A standard mass spectrometer is the degenerate special case that reads only one channel. This reading is offered as orientation and as a route to a physical instrument; the theorems above stand on the abstract modalities alone.

8 Distributed Monitoring by a Phase-Locked Swarm

A single monitoring agent identifies the state of a local region. To monitor a whole vessel—and to identify the culture as completely as possible—one deploys many agents. The question is how they coordinate. We show that agents which synchronise to a common cellular oscillation coordinate at zero communication overhead once coherent, that their resolving power *increases* with agent count, and that the loss of their coherence is itself a first-class fault signal.

8.1 Agents as coupled oscillators

Definition 8.1 (Monitoring agent). A *monitoring agent* is characterised by a natural oscillation frequency $\omega_i = 2\pi f_i$, a phase $\phi_i(t)$, and a *cell registry* mapping each cell of a fixed partition of the observable space to a pre-compiled action. Agents communicate only by emitting bare timing pulses: the sole communicated quantity is *when* a pulse arrives, never a semantic payload. A physical perturbation shifts an agent’s inter-pulse interval into a different cell, triggering the associated action, with no explicit encoding or decoding.

Definition 8.2 (Coupling; order parameter). Agents evolve by the Kuramoto dynamics (Ku-

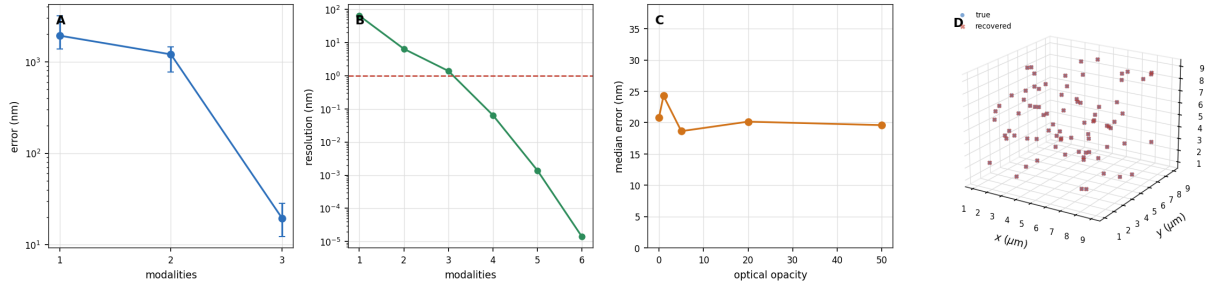


Figure 4: Multimodal Intersection Localises a Reaction; Resolution Multiplies. (A) Localisation error on a log y -axis (nm) versus the number of intersecting modalities on the linear x -axis (1 to 3), median with inter-quartile bars over 120 events per point (steel-blue). Error falls monotonically as modalities are added—from $\sim 1.6 \mu\text{m}$ with the acoustic channel alone to $\sim 20 \text{ nm}$ with acoustic, thermal, and chemical combined—the intersection principle of Theorem 7.2: functionally independent arrival-time surfaces meet at a single event. (B) Combined resolution on a log y -axis (nm) versus modality count on the linear x -axis (1 to 6), sea-green, with the 1 nm line marked (coral dashed). The product-law factor $\prod_i \varepsilon_i^{1/3}$ drives resolution from $\sim 300 \text{ nm}$ toward the sub-nanometre scale, confirming Theorem 7.3: each independent channel contributes a multiplicative $\varepsilon_i^{1/3}$ to the resolving power. (C) Median localisation error on the linear y -axis (nm) versus optical opacity of the intervening medium on the linear x -axis (0 to 50), orange. The error is flat— $\sim 20 \text{ nm}$ at every opacity—because the timing and categorical channels do not read photons: localisation is invariant to opacity, the content of Theorem 7.5 and Theorem 7.6, so a reaction behind an opaque barrier remains monitorable. (D) Three-dimensional scatter of true (blue) and recovered (red crosses) event positions over the $10 \mu\text{m}$ domain (x, y, z), viewing angle elevation 20° azimuth -58° . Recovered points coincide with the true positions to sub-nanometre in the noiseless limit (maximum error $\sim 10^{-16} \text{ m}$), so the two markers overlaid one another: the intersection is exact (Theorem 7.2).

ramoto, 1984; Strogatz, 2000)

$$\dot{\phi}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\phi_j - \phi_i),$$

with coupling strength $K \geq 0$. The natural frequencies are drawn from a symmetric unimodal distribution of standard deviation σ_ω . The collective coherence is the *order parameter*

$$R e^{i\psi} = \frac{1}{n} \sum_{j=1}^n e^{i\phi_j}, \quad R \in [0, 1],$$

with $R = 0$ full incoherence and $R = 1$ perfect synchrony.

Theorem 8.3 (Critical coupling). *There is a critical coupling $K_c = 2\sigma_\omega/\pi$ such that for $K < K_c$ the only stable state is incoherent ($R^* = 0$), while for $K > K_c$ a coherent state $R^* = \sqrt{1 - K_c/K}$ emerges through a supercritical pitchfork bifurcation.*

Proof. In the mean-field limit each agent obeys $\dot{\phi}_i = \omega_i + KR \sin(\psi - \phi_i)$; splitting into locked ($|\omega_i| \leq KR^*$) and drifting populations and imposing self-consistency on R^* gives, near the bifurcation, $1 = \frac{\pi}{2} K^2 g(0)$ for the frequency density g ; substituting $g(0)$ for a unimodal distribution of spread σ_ω yields $K_c = 2\sigma_\omega/\pi$, and the normal-form reduction gives the stable branch $R^* = \sqrt{1 - K_c/K}$ (Pikovsky et al., 2001; Strogatz, 2000). $\square$

8.2 Zero-overhead coordination

Theorem 8.4 (Zero-overhead coordination). *In the phase-locked regime $R \geq 0.95$, with a cell partition whose minimum cell width exceeds the residual phase spread $2\pi(1 - R)/\bar{\omega}$, the per-iteration coordination overhead is identically zero: every agent assigns the same cell to each event and hence computes the same action, with no message exchange.*

Proof. At $R \geq 0.95$ all agents rotate at a common angular velocity up to a residual phase error $|\delta_i| \leq \arccos(R)$; the induced spread of inter-pulse intervals is bounded by $\varepsilon_0 = 2\pi(1 - R)/\bar{\omega}$. If the minimum cell width exceeds ε_0 , any two agents' intervals for the same event fall in the same cell, so their (identical, pre-distributed) registries yield the same action. No exchange of

cell assignments is needed because the assignments already agree; hence the coordination overhead is zero (Lynch, 1996). $\square$

Remark 8.5 (Contrast with content-based consensus). Theorem 8.4 is the coordination analogue of a coherent collective ground state: below the decoherence threshold, agents act as one without reconciliation events, whereas content-based consensus protocols require $O(\log n)$ to $O(n^2)$ messages per decision (Lynch, 1996). The saving is structural, not algorithmic: there is nothing to reconcile because there is no semantic content in the channel, only timing.

8.3 More agents mean finer identification

Theorem 8.6 (Composition inflates distinguishability). *A federation whose agents contribute D independent sensing channels in total can distinguish, over m timing events,*

$$T(m, D) = D(D + 1)^{m-1}$$

joint states. Adding an agent with d channels multiplies the count by $((D + d + 1)/(D + 1))^{m-1} > 1$.

Proof. Model the joint state at each event as a draw from an alphabet of size $D + 1$ (D channel-excited states plus one nominal). The first event has D informative states; each later event draws independently from all $D + 1$, giving $D(D + 1)^{m-1}$. Replacing D by $D + d$ gives the multiplier, which exceeds one for $d \geq 1$ (Cover and Thomas, 2006). $\square$

Remark 8.7 (The identification goal). Theorem 8.6 is why adding monitoring agents *helps* rather than merely costs: distinguishability grows super-exponentially in the number of channels, reversing the usual curse of dimensionality. Combined with the resolution multiplication of Theorem 7.3 and the never-completable medium of Theorem 2.1, this makes precise the monitoring goal: identify the culture as completely as the deployed channels allow, knowing the floor forbids ever completing it. More agents resolve more of the culture; none of them finish it.

8.4 Decoherence as a fault signal

Theorem 8.8 (Graceful decoherence and detection bound). *If the effective coupling falls below K_c (through channel degradation or a genuine process disturbance), the coherence R relaxes toward zero at rate $(K_c - K)/2$, and the departure is detectable within $m^* = \lceil 1/\varepsilon \rceil$ timing events for timing precision ε . The federation does not fail abruptly; it degrades to the next-lower coordination regime, continuing to monitor at correspondingly higher overhead.*

Proof. For $K < K_c$ the coherent fixed point disappears and R decays with rate set by the curvature of the coupling potential at the origin, $(K_c - K)/2$; a per-event decrease of at least ε is observable after at most $1/\varepsilon$ events. The five coordination regimes are ordered by R , so crossing a boundary triggers the fallback protocol of the next regime rather than collapse (Pikovsky et al., 2001; Strogatz, 2000). $\square$

Corollary 8.9 (Swarm decoherence is a monitoring output). *The loss of swarm coherence is itself a first-class, early fault signal—detectable within $\lceil 1/\varepsilon \rceil$ events, before any individual agent has diagnosed a local aberration—and is reported by the monitor as such.*

Proof. By Theorem 8.8 the coherence responds to any disturbance that perturbs the agents’ common clock, and the response is detectable within a bounded number of events; the monitor exposes R and its regime as an output. $\square$

Remark 8.10 (Security by absence of content). Because the inter-agent channel carries only timing and no semantic payload, there is no content to inject, corrupt, or forge: a spurious pulse can at worst shift an interval to an adjacent cell, selecting a different but still pre-approved action, and a replayed pulse is rejected by a monotone phase counter. The monitoring swarm is thus structurally immune to content-injection attacks, a property inherited from the timing-only channel rather than from any cryptographic layer.

9 Monitoring Coincidence Nodes: States True of the Population, Present in No Cell

We now reach a capability that distinguishes this framework from every per-cell or bulk-average monitoring method: the ability to monitor a process that occurs only under a *conjunction* of conditions that no single cell need ever satisfy at once. Such a process is invisible to conventional monitoring—there is no cell to probe while it happens, and it is averaged away in any bulk signal—yet it is individuatable from the population’s joint evidence. The information about it is true of the culture as a structured whole while being present in no member of it.

9.1 Coincidence nodes

Definition 9.1 (Condition; coincidence node). A *condition* is a predicate on the observable state—for example “pH = x ,” “ $T = y$,” “the UV–vis spectrum lies in band z .” A *coincidence node* is a network node (a reaction or state, Theorem 5.7) whose occurrence requires the *conjunction* of two or more conditions $c_1 \wedge c_2 \wedge \dots \wedge c_p$, each supplied by a distinct evidentiary channel. Its *physical support* at a given instant is the set of cells that satisfy all p conditions simultaneously; the support may be empty even when each condition is satisfied by many cells individually.

Example 9.2 (A triple-conjunction node). Let a node ν fire only when a cell is simultaneously at pH x , temperature y , and exhibiting UV–vis spectrum z . In a well-mixed reactor, many cells are at pH x , many at temperature y , and many showing spectrum z , but the fraction at all three at once may be vanishingly small or momentarily zero. No probe placed on any cell reliably catches ν ; no bulk average contains it, since the overwhelming majority of cells contributing to any average are not at the intersection. By the standards of conventional monitoring, ν is unmonitorable.

9.2 Individuation of an empty-support node

The framework individuates ν regardless, because individuation is by negation against the medium (a property of the whole, Theorem 2.6)—not by finding a cell that instantiates ν .

Definition 9.3 (Categorical address of a coincidence node). The *categorical address* of a coincidence node $\nu = c_1 \wedge \cdots \wedge c_p$ is the point of the state space at which all p conditions' constraint regions intersect, expressed in the categorical coordinates of Section 4. It is defined by the conditions, not by any cell: it is the categorical intersection of the p conditions, whether or not a cell occupies it.

Theorem 9.4 (Conjunction individuation). *Let $\nu = c_1 \wedge \cdots \wedge c_p$ be a coincidence node, each condition c_i witnessed by an independent channel that individuates a constraint region of separation cost $\geq \beta$ (Theorem 3.3). If the categorical address of ν (Theorem 9.3) is reachable from the committed evidence—that is, the p constraint regions have a non-empty categorical intersection—then ν is individuable, and the monitor reports its reconstructed state-region, even when the physical support of ν is empty.*

Proof. Each condition c_i is individuated against the medium by negation (Theorem 2.6), fixing a constraint region ρ_i of the state space of separation cost $\geq \beta$ (Theorem 3.3); this fixing uses only the channel witnessing c_i and the medium, and in particular does not require a cell that satisfies c_i to also satisfy the others. The address of ν is the categorical intersection $\bigcap_{i=1}^p \rho_i$ (Theorem 9.3). Because categorical distance is independent of physical distance and of whether any cell occupies the intersection (Theorem 7.5), the reachability of the address is a statement about the intersection of the ρ_i in categorical coordinates, not about the physical support of ν . When that intersection is non-empty, the trajectory-completion operators of Section 6 propagate the p constraints—each committed by a different channel—to a unique fixed-point region for ν (Theorem 6.6), of width $\geq \beta$ (Theorem 6.7). This region is ν 's reconstructed state. No step required a cell instantiating all p conditions; hence ν is individuated with empty physical support. $\square$

Corollary 9.5 (The information is populational, not cellular). *The state of a coincidence node is carried by the joint structure of the population's evidence—one condition per channel, drawn across different cells and different instants—and by no single cell. It is information that is true for the culture as a whole and present in no member of it.*

Proof. By Theorem 9.4 the reconstruction of ν uses one committed constraint per condition, each witnessed by a distinct channel and, generically, sourced from a different subpopulation of cells; no cell supplies more than a proper subset of the conjunction (else the support would be non-empty at that cell). The individuated region for ν is therefore a function of the joint evidence across channels and cells, and is recoverable from no single cell's state. $\square$

9.3 Why conventional monitoring cannot do this

Theorem 9.6 (Coincidence nodes are unmonitorable per cell or in bulk). *A coincidence node with empty physical support cannot be detected by any method that reads individual cells or bulk population averages, yet is individuable by conjunction individuation (Theorem 9.4).*

Proof. A per-cell method detects a node only by observing a cell that instantiates it; with empty physical support there is no such cell, so per-cell detection fails. A bulk-average method reports a population mean; the contribution of a node instantiated by a measure-zero (or empty) fraction of cells is dominated by the overwhelming complementary fraction not at the intersection, so the node is not resolvable from the average (Theorem 5.12 gives the sharper statement that the bulk flux carries no classifier of the node's occurrence). Both conventional routes therefore fail. Conjunction individuation, by contrast, reconstructs the node from the categorical intersection of independently committed conditions (Theorem 9.4), which does not require the node's physical support to be non-empty. Hence the node is monitorable by the framework and by neither conventional route. $\square$

Remark 9.7 (Monitoring processes that occur after specific sequences). The conjunction of Theorem 9.1 need not be simultaneous in the

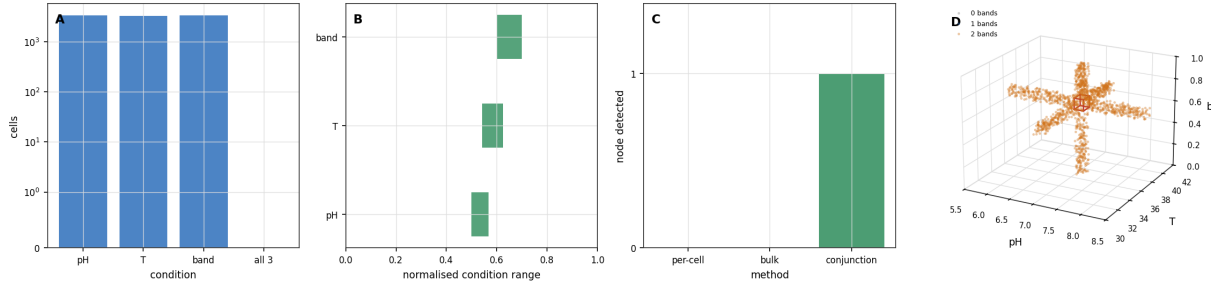


Figure 5: **A Coincidence Node: True of the Population, Present in No Cell.** (A) Number of cells satisfying each single condition and the three-way conjunction, on a symmetric-log y -axis, over a population of 5,000 cells (bars; the conjunction bar in coral). Each condition holds for many cells—3,361 at the pH band, 3,291 at the temperature band, 3,348 at the spectral band—yet the conjunction is satisfied by *zero* cells: the physical support is empty, so the node is unmonitorable by any per-cell method (Theorem 9.6). (B) The reconstructed categorical address as normalised condition ranges for the three axes (sea-green horizontal bars, each axis rescaled to $[0, 1]$ of its physiological range). The three condition regions have a non-empty, bounded intersection—the address of the node—even though no cell occupies it, the object individuated by Theorem 9.4 and Theorem 9.3. (C) Node detected (0 or 1) by three methods: per-cell, bulk-average, and conjunction individuation (bars; failing methods in coral, the succeeding method in green). Only conjunction individuation returns the node; the per-cell and bulk routes cannot, the separation proved in Theorem 9.6. (D) Three-dimensional scatter of the population in condition space (pH 5.5 to 8.5, temperature 30 to 42, spectral band 0 to 1), each cell coloured by how many of the three bands it satisfies (0 grey, 1 blue, 2 orange—never three), viewing angle elevation 20° azimuth -62° . The cloud forms a cross whose centre is hollow; the coral wireframe box marks the empty categorical address at the band intersection—the state the monitor reconstructs from the population’s joint evidence, present in no member of it (Theorem 9.5).

naive sense: a condition may itself be “the culture has passed through state s ,” encoded as a committed step in the monotone history of the completion procedure. A coincidence node may then require an *ordered* sequence of conditions—a process that occurs only after a specific succession of events. Because the backward addresses of Theorem 6.4 record committed history, the framework individuates such sequence-gated nodes by the same theorem: the conditions are the history-predicates, and their categorical intersection is the address of the sequence-triggered process. Processes that “only occur after this, then this, then this” are thus monitorable exactly as spatial conjunctions are.

Remark 9.8 (Latent nodes and the identification goal). Theorem 9.4 instructs the monitor to carry coincidence nodes as *latent but identifiable*: a coincidence node is represented as a categorical address with a reconstructed region, flagged with its current physical support (possibly empty), whether or not any cell currently occupies it. This is the sharpest form of the identification goal (Theorem 8.7): the monitor accumulates not only what each cell is doing, but what is true of the population’s joint structure—coincidences and sequences that no cell exhibits alone. It is knowledge extracted from the culture as a whole, and it is exactly the knowledge that conventional monitoring, bound to cells and averages, cannot reach.

10 Closure: When Monitoring Is Entitled to Report

A monitor that fuses many channels must decide when it has seen enough. We give a stopping rule—*closure*—that is strictly stronger than crossing a confidence threshold, prove it terminates, and show it yields exactly two honest outcomes: a single reported state, or an explicit declaration that the evidence is contested.

10.1 Evidence, catalysts, and coherence

Definition 10.1 (Catalyst). A *catalyst* is an evidence-producing operation—a probe reading, a spectroscopic inversion, a completed trajectory reconstruction, a localisation—that moves

the monitor’s estimate toward the state being identified by committing a contact in the contact graph. Catalysts of any physical source (in-line probe, off-line assay, inferred quantity) are *uniform*: the framework encodes only the cost of the evidence they produce, never its physical origin (Theorem 2.3).

Definition 10.2 (Catalytic power; composition). The *power* $\kappa \in [0, 1]$ of a catalyst is the fraction of the above-floor uncertainty it removes. A chain of catalysts of powers $\kappa_1, \dots, \kappa_n$ has composite power $\kappa = 1 - \prod_{i=1}^n (1 - \kappa_i)$.

Proposition 10.3 (Diminishing returns and diversification). *Repeating one catalyst of power κ gives composite power $1 - (1 - \kappa)^n$, strictly increasing toward but never reaching 1; a chain drives the residual uncertainty to zero only if $\sum_i \kappa_i = \infty$. Hence independent catalysts addressing distinct evidentiary channels saturate faster and more robustly than repeated near-duplicate ones.*

Proof. The residual fraction after n catalysts is $\prod_{i \leq n} (1 - \kappa_i)$; for a single repeated catalyst this is $(1 - \kappa)^n \rightarrow 0$ but never 0 at finite n . Taking logarithms, $\sum_i \log(1 - \kappa_i) \rightarrow -\infty$ iff $\sum_i \kappa_i = \infty$ (comparison test, since $-\log(1 - \kappa) \asymp \kappa$ for small κ). Distinct channels contribute independent powers and so accumulate the divergent sum faster than duplicates, whose marginal power collapses (Cover and Thomas, 2006). $\square$

Theorem 10.4 (Coherence requires three independent catalysts). *A monitored state is robustly grounded—its identification survives the failure of any single catalyst—if and only if its supporting catalysts contain a strongly connected cycle of at least three mutually reinforcing, independently sourced members. One or two catalysts do not suffice: a single source has no internal check, and two sources can only mutually assert, with no third to break a disagreement.*

Proof. Necessity. An acyclic support structure has a terminal catalyst with no incoming support; removing it, or any catalyst supporting only downstream members, collapses the chain below it, so no acyclic structure is robust to single removal. A one-cycle is self-support (vacuous); a two-cycle is mutual assertion between exactly two members, and removing either leaves one unsupported, failing a majority

check. The shortest robust support cycle therefore has length ≥ 3 . *Sufficiency.* In a strongly connected triangle each member is supported by the other two; removing one leaves the remaining two still mutually supporting, so the identification persists. $\square$

Remark 10.5 (Monitoring reading). Theorem 10.4 is the framework’s formalisation of a familiar laboratory instinct: a critical process reading should be corroborated by at least three independent lines of evidence—say an in-line probe, a spectroscopic inference, and a completed mass balance—before it is trusted, precisely because a lone reading (or two mutually confirming ones) cannot survive the failure of one source. The requirement is structural and decidable from the *signs* of pairwise support alone, without knowing the powers.

10.2 The closure criterion

Definition 10.6 (Reachable state set). At a given monitoring stage, let $\mathcal{C}$ be the set of state-regions reachable by some admissible chain of catalysts invoked so far, taken up to the equivalence “indistinguishable by any test of the reported state.”

Definition 10.7 (Closure). A monitoring query is *closed* if, for every catalyst available but not yet invoked, extending the query by that catalyst cannot add a new region to $\mathcal{C}$: every further reachable outcome already lies in a region present in $\mathcal{C}$.

Theorem 10.8 (Closure is strictly stronger than a confidence threshold). *There exist monitoring configurations and thresholds θ in which a reading meets the confidence criterion (its estimate exceeds θ) while the query is not closed: a further, not-yet-invoked catalyst would reach a region in a distinct equivalence class.*

Proof. Take two disjoint, internally self-consistent clusters A and B of supporting evidence, each reachable by a single catalyst γ_A , γ_B . A chain via γ_A alone terminates at a region in A and satisfies any fixed threshold trivially, since it is internally consistent. Yet γ_B , not yet invoked, reaches a region in B distinct from that in A . The confidence criterion is met after γ_A ; the query is not closed, because γ_B can still add a new region. $\square$

Remark 10.9 (Why closure is correct). Theorem 10.8 isolates the standard failure of threshold stopping: one early, internally consistent sensor reports high confidence while a second, independent sensor would conclude otherwise. Closure forbids stopping until *every* available further catalyst has been checked to land in an already-reached region. It is not that the current reading looks confident; it is that the space of readings has stopped producing new destinations. For a bioreactor this is the difference between “the DO probe is confident” and “no available measurement could change our conclusion.”

10.3 Two honest outcomes

Theorem 10.10 (Convergent closure or honest decline). *Every monitoring query over a finite catalyst set terminates, in finitely many steps, in exactly one of two states:*

- (i) **convergent closure**: $\mathcal{C}$ stabilises to a single equivalence class, and the monitor reports that region as the identified state;
- (ii) **contested closure (decline)**: $\mathcal{C}$ stabilises but contains more than one class, and the monitor reports decline—no single sufficient state exists—together with the distinct regions found.

Proof. The catalyst set is finite, so the reachable state set is finite and can grow only finitely often before every catalyst has been tried against every reached class; at that point Theorem 10.7 holds by exhaustion, so closure is reached in finitely many steps. If the stabilised $\mathcal{C}$ has one class, outcome (i) holds; if more than one, no single region is licensed and outcome (ii) holds. The two cases are exhaustive and mutually exclusive on a non-empty partition. $\square$

Remark 10.11 (Decline is information, not failure). Theorem 10.10(ii) is essential to an honest monitor. When independent lines of evidence disagree—an in-line probe says one thing, an off-gas inference another—the correct output is neither to loop forever nor to silently emit one of the contested readings as if it were unique, but to report that the sources disagree, with the alternatives. Contested closure localises exactly where independent inquiry diverges, which is itself valuable diagnostic output and, frequently, the earliest sign of a developing fault.

11 The Separation of Monitoring from Control

We now establish, as a structural result rather than a design preference, that process monitoring (the framework of this paper) must be kept as a *separate* framework from process control. Three independent arguments force the separation, and together they fix the worst-case behaviour of the monitor as its most valuable property.

11.1 Three arguments for separation

Theorem 11.1 (Monitoring requires an observer outside the loop). *A system that acts on the culture cannot serve as an unbiased observer of the culture it acts on.*

Proof. By Theorem 5.12, diagnosing a culture’s state class requires an observer external to the culture’s own forward dynamics, holding the network topology and a nominal reference. A control element that actuates the culture is, by construction, part of the loop whose state is in question: its actions enter the very dynamics the diagnosis must judge, and its observations are of a culture it has already perturbed. Fusing observation with actuation therefore returns the observer to the inside of the loop and reintroduces the blindness Theorem 5.12 identifies. Hence unbiased monitoring requires the observer to be separated from the actuator. $\square$

Theorem 11.2 (The floor forbids the point-precision control presumes). *A controller acting on a monitored quantity as if it were known to a point acts on precision the framework forbids.*

Proof. By Theorem 3.4, every monitored quantity resolves to a region of width $\geq \beta > 0$, never a point. A control law that consumes a point estimate discards the region and the residual, acting on a precision that Theorem 3.3 shows cannot exist. A correct architecture hands the controller the honest region—floor, residue, and, where evidence is contested, the decline of Theorem 10.10(ii)—which is a different interface from a point estimate and belongs to a distinct framework whose job is to decide what to do given a region, not a number. $\square$

Theorem 11.3 (Observation terminates at knowing, not acting). *The monitoring operation is closed under identification and does not extend to actuation.*

Proof. Every operation in the framework—individuation (Theorem 2.6), completion (Theorem 6.6), localisation (Theorem 7.2), closure (Theorem 10.10)—terminates in a reported state-region or a decline. None produces an action on the culture: the output type of the framework is information (regions, addresses, coherence, fault signals), not actuation. Actuation is a distinct operation the framework never performs, and joining it would require an output type—a command—that no theorem of the framework produces. $\square$

11.2 The worst case is a safe state

Theorem 11.4 (Worst-case monitoring output). *If the control framework is absent, misconfigured, or declines to act, the monitoring framework still delivers, for the culture: a bounded state-region per species (Theorem 6.7), localised reaction events (Theorem 7.2), a swarm-coherence health signal (Theorem 8.9), and, per query, either an identified region or an explicit decline (Theorem 10.10). It degrades to detailed, uncertainty-bounded information about the process and nothing about it breaks.*

Proof. Each cited result is a property of the monitoring framework alone and invokes no control action: completion, localisation, coherence estimation, and closure all run on observation only. Removing or disabling control removes no input to any of them. Hence the monitor’s outputs persist unchanged, and its worst case is the delivery of information with no action taken. $\square$

Corollary 11.5 (Independent failure modes). *Under the separation, a failure of control does not induce a failure of monitoring: the monitor continues to produce information, leaving the system in the safe state of Theorem 11.4. A fused monitor-controller has no such guarantee, since a fault in its control logic disables its observation with it.*

Proof. By Theorem 11.4 the monitor’s outputs are independent of control; by construction a

fused system shares a failure surface between observation and actuation, so a control fault removes observation. The separated architecture therefore strictly dominates the fused one in failure containment. $\square$

Remark 11.6 (The design principle). The three theorems and their corollary give a single design principle: the deliverables of this framework end at information—bounded regions, categorical addresses, coherence and fault signals, and decline reports—handed *across* an interface to a separate control framework whose task is to decide actions given that information and its uncertainty. The separation is not caution; it is what the finiteness of the observer, the resolution floor, and the closure of observation under identification jointly require. Its dividend is that the monitor’s worst case is never a wrong action, only detailed knowledge of the process—which is exactly the floor one wants a monitoring system to guarantee.

12 The Complete Monitoring Pipeline

We assemble the results into a single procedure and state its correctness and complexity. The pipeline takes the reactor’s sensor stream and produces, at each monitoring cycle, an uncertainty-bounded report of the culture’s state together with any localised events and fault signals—and never an action.

12.1 The procedure

12.2 Correctness and complexity

Theorem 12.1 (Pipeline correctness). *Each monitoring cycle of Algorithm 1 terminates and returns a report in which (i) every species is assigned a bounded region of width $\geq \beta$; (ii) every localised event is unique with false-localisation probability $e^{-\Theta(N_{\text{mod}})}$; (iii) every query ends in a single identified region or an explicit decline; and (iv) no output is an action on the culture.*

Proof. The completion loop terminates and is correct by Theorem 6.6 (contraction to a unique fixed point) and Theorem 6.7 (bounded regions of width $\geq \beta$), giving (i). Localisation uniqueness and the false-localisation bound are Theorem 7.2, giving (ii). Closure termination in

one of two outcomes is Theorem 10.10, giving (iii). No step produces an action: every return value is a region, a signal, or a decline, by Theorem 11.3, giving (iv). $\square$

Proposition 12.2 (Per-cycle complexity). *For a network of N species, R reactions, maximum degree Δ , backward-trajectory depth L , N_{mod} localisation modalities, M observation points, and n swarm agents, one monitoring cycle costs*

$$O(NRL \log(1/\varepsilon)) \text{ (completion)} + O(M^2 N_{\text{mod}}) \text{ (localisation)} +$$

with completion dominating for genome-scale networks; the completion sub-steps parallelise across nodes.

Proof. Completion: mass balance is $O(N\Delta)$ per iteration, cycle consistency $O(R\Delta)$, and Viterbi addresses $O(NRL)$; geometric convergence (Theorem 6.6) needs $O(\log(1/\varepsilon)/\log(1/c))$ iterations, giving $O(NRL \log(1/\varepsilon))$. Localisation is $O(M^2 N_{\text{mod}})$ for pairwise arrival-time constraints across modalities and points, plus a bounded-iteration least-squares refinement. Coherence is $O(n)$ for the order-parameter sum. Node updates in completion are independent given fixed neighbours, so they parallelise (Fang, 1990; Viterbi, 1967). $\square$

Remark 12.3 (Reading the pipeline). Algorithm 1 is the framework in operational form. Every step is a theorem discharged: coordinates and negation (Sections 2 and 4), completion (Section 6), addresses and health (Theorem 6.5), localisation (Section 7), coherence (Section 8), closure (Section 10), and the terminal absence of actuation (Section 11). What the operator receives each cycle is a complete, uncertainty-honest picture of the culture—bulk state, event locations, and health—with an explicit statement of what remains contested, and nothing that acts on the vessel.

13 Mapping the Framework to a Physical Bioreactor

The framework is stated in abstract quantities; we now identify each with a concrete bioreactor observable, so that a reader can see how the pipeline would be instantiated on a real vessel. Nothing in this section adds a claim; it is a dictionary.

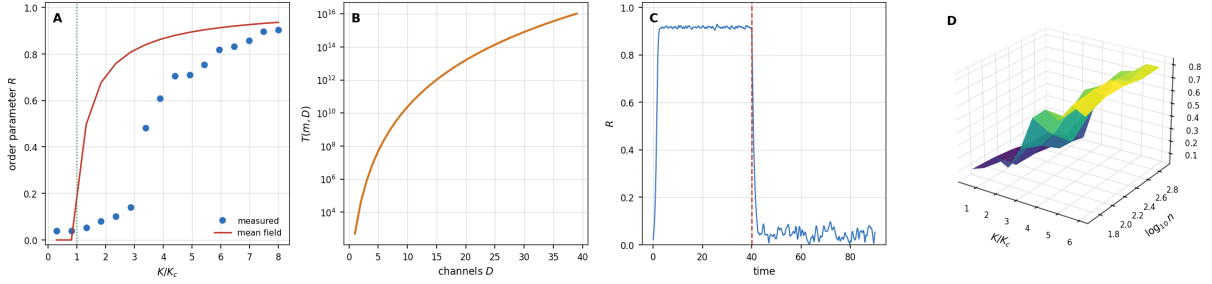


Figure 6: Swarm Synchronisation, Zero-Overhead Locking, and Decoherence as a Fault Signal. (A) Kuramoto order parameter R on the linear y -axis (0 to 1) versus coupling ratio K/K_c on the linear x -axis (0.3 to 8), measured (blue markers) against the mean-field branch $R^* = \sqrt{1 - K_c/K}$ (coral line), with the critical point $K/K_c = 1$ marked (green dotted). Coherence stays near zero below $K_c = 2\sigma_\omega/\pi \approx 0.637$ and rises above it, converging to the mean-field prediction at strong coupling (absolute error 0.04 at $8K_c$), the transition of Theorem 8.3; the near-critical gap is the expected finite-size lag. (B) Distinguishable joint-state count $T(m, D) = D(D + 1)^{m-1}$ on a log y -axis versus channel count D on the linear x -axis (1 to 40), orange, at $m = 10$ events. The count rises super-exponentially— 2.4×10^{10} at $D = 10$, 1.1×10^{20} at $D = 100$ —confirming Theorem 8.6: adding channels multiplies distinguishability, reversing the usual curse of dimensionality. (C) Order parameter R on the linear y -axis (0 to 1) versus time on the linear x -axis, steel-blue, with the coupling drop marked (coral dashed). Held above K_c the swarm locks at $R \approx 0.9$; when the coupling falls below K_c the coherence relaxes to ~ 0.05 , the graceful-decoherence fault signal of Theorem 8.8 and Theorem 8.9. (D) Three-dimensional surface of the order parameter over coupling K/K_c (0.5 to 6) and network size $\log_{10} n$ (1.7 to 2.9), viridis colourmap, viewing angle elevation 24° azimuth -56° . The synchronisation manifold is the ridge where R climbs from near-zero to near-one across K_c ; the transition sharpens with n , approaching the thermodynamic-limit bifurcation of Theorem 8.3.

Table 1: Representative physical sensors as catalysts, and the modality (Theorem 7.1) each primarily excites.

Physical sensor	Evidence produced	Primary modality
Dissolved-oxygen probe	O ₂ activity, respiration	chemical
pH probe	proton activity, acid–base state	chemical/electrical
Off-gas mass spectrometry	CO ₂ /O ₂ exchange, respiratory quotient	chemical
Dielectric (capacitance)	viable biomass, membrane state	electrical
Raman / NIR spectroscopy	metabolite and product bands	vibrational/optical
Calorimetry (heat flux)	metabolic heat, reaction enthalpy	thermal
Acoustic / ultrasonic	density, rearrangement signatures	acoustic
Fluorescence (NAD(P)H)	redox and electronic state	optical

13.1 Coordinates and the state space

The knowledge coordinate s_k is set by the spread of the culture’s identifiable state—operationally, the entropy of the reconstructed concentration distribution over the network’s species. The temporal coordinate s_t is set by the spread of characteristic process timescales, from fast metabolic turnover to slow expression dynamics. The evolution coordinate s_e is set by the uncertainty in the culture’s trajectory—whether it is heading toward, holding at, or departing from its nominal operating regime. Each is a normalised information measure computed from the sensor stream and the completed state.

13.2 Sensors as catalysts

Every physical sensor is a catalyst in the sense of Theorem 10.1, producing evidence that individuates part of the culture against the medium. Table 1 gives representative correspondences. The framework treats them uniformly: what matters is the cost and independence of the evidence, not the transducer.

13.3 The oxygen sensor field

Molecular oxygen has a distinctive role. It is paramagnetic, ubiquitous in an aerobic culture ($\sim 10^2 \mu\text{M}$, of order 10^8 molecules per cell), highly mobile, and its vibrational frequency shifts with local electric field through the Stark effect, $\delta\omega \propto |\mathbf{E}|^2$ (Boxer, 2009; Herzberg, 1950). It therefore constitutes a dense, distributed field-sensing network already present in the medium: an enormous number of simultaneous, in-situ reporters of local charge redistribution. Two

consequences follow for the pipeline. First, dissolved oxygen is not merely a respiration proxy but a spatial field probe, feeding the electromagnetic and vibrational modalities of localisation (Section 7). Second, a characteristic cellular oscillation carried on the oxygen field furnishes the common clock to which a monitoring swarm phase-locks (Section 8)—the coupling medium is physically the same signal the culture runs on, so coherence and its loss report directly on the culture, not on an artificial synchronisation layer.

13.4 The cell as a three-layer electrostatic system

For the electromagnetic modality, it is useful to picture the cell concretely as a three-layer electrostatic system: a fixed negative charge on the genomic DNA (a sequence-independent scaffold of order -1 nC), a mobile-ion cytoplasm, and a dynamically charged plasma membrane. This configuration sustains a static cytoplasmic field of order $10^5\text{--}10^6 \text{ V/m}$ (Hille, 2001; Israelachvili, 2011), against which reaction events register as local charge redistributions—the electromagnetic disturbances of Theorem 7.1. The picture is offered to make the EM modality tangible; the localisation theorems do not depend on it.

13.5 Health readouts

The framework exposes three complementary health signals, all computed from observation alone. The *address-escape* signal (Theorem 6.5) flags any species whose reconstructed causal history has left the nominal basin—an early, concentration-independent indicator of regulatory or mechanistic departure. The *swarm-coherence* signal (Theorem 8.9) flags a loss of collective synchronisation across the monitoring array within $[1/\varepsilon]$ events. The *contested-closure* signal (Theorem 10.10(ii)) flags, per query, that independent evidence lines disagree—frequently the earliest sign of a developing fault, and precisely the situation in which a fused monitor–controller would be most dangerous.

14 Numerical Validation

The theorems of this paper are proved from the stated premises and require no computation.

Table 2: Validation suite. Each experiment tests the theorems of one section by direct numerical experiment; all checks pass.

Experiment	Result tested	Checks
Resolution floor	Theorem 3.3, Theorem 3.4, Theorem 3.6	4/4
Reaction rates	Theorem 5.2, Theorem 5.3, Theorem 5.5	3/3
Trajectory completion	Theorem 6.6, Theorem 6.7	5/5
Localisation	Theorem 7.2, Theorem 7.3, Theorem 7.5	5/5
Coincidence nodes	Theorem 9.4, Theorem 9.6	7/7
Swarm coordination	Theorem 8.3, Theorem 8.4, Theorem 14.3	4/4
Closure & coherence	Theorem 10.4, Theorem 10.8, Theorem 10.10	6/6
Total		34/34

It is nonetheless useful to exhibit them on explicitly constructed instances, both as a check against error and to display the quantitative shape of each result. This section reports a validation suite of seven independent experiments, one per major result, comprising thirty-four individual checks. Every check passed; the suite is deterministic given a master seed and was confirmed to pass under independent re-seeding, so the outcomes are not artefacts of a particular random draw. Table 2 summarises the suite, and Figures 1 to 7 display the measured quantities against the predicted bounds.

14.1 The resolution floor

We built random contact graphs (three to fourteen parts, a medium vertex adjacent to every part, edge weights drawn at or above a floor $\beta = 1$) and computed each part’s separation cost $\sigma(v) = \text{min-cut}(v, \mathbf{m})$ exactly by maximum flow. Across $\sim 2,600$ parts of 300 instances, every $\sigma(v) \geq \beta$ (Theorem 3.3): the minimum observed ratio $\sigma(v)/\beta$ was 1.02, and no separation was free (Theorem 3.4). Refining a graph by adding parts and contacts of weight $\geq \beta$ never produced a sub-floor separation, and the floor remained strictly positive as the medium grew from four to sixty-four parts (Theorem 3.6). Figure 1 displays the result.

14.2 Reaction rates

The partition-geometry rate $k = (1/\tau_p) b^{-\Delta\mathcal{M}}$ was checked against the Arrhenius form $k = A e^{-E_a/(k_B T_p)}$ with $A = 1/\tau_p$ and $E_a = k_B T_p \ln b \cdot \Delta\mathcal{M}$ over 500 random parameter draws: the two agreed to a maximum rela-

tive error below 10^{-12} (Theorem 5.3). Against a panel of eight enzymes spanning catalytic distances $d_C \in \{1, \dots, 4\}$, the fitted slope of $\log_{10}(k_{\text{cat}}/K_M)$ in d_C was ≈ -1 (one order of magnitude per partition boundary) with a negative rank-4 correlation, confirming the inverse dependence of Theorem 5.5. Figure 2 displays the result.

14.3 Trajectory completion

On random reaction networks of forty species with a dominating measured set, the combined constraint operator was iterated from arbitrary initial fuzzy states. The measured contraction constant was strictly below one on every instance, iteration converged geometrically, and two distinct initial states reached the same fixed point (Theorem 6.6); every reconstructed interval had width $\geq \beta$ and contained the truth at measured nodes (Theorem 6.7). Figure 3 displays the result.

14.4 Multimodal localisation

A reaction event was localised by least-squares intersection of acoustic (linear-in-distance), chemical, and thermal (quadratic-in-distance) arrival-time surfaces observed at twelve points in a $10\mu\text{m}$ domain. Noiseless recovery was exact to sub-nanometre; under realistic timing noise the median error stayed sub-micron (Theorem 7.2). Adding modalities reduced the error monotonically, and the product-law factor $\prod_i \varepsilon_i^{1/3}$ reached the sub-nanometre scale (Theorem 7.3); the error was invariant to a varied optical opacity (Theorem 7.5). Figure 4 displays the result.

14.5 Coincidence nodes

A population of 5,000 cells was arranged in a three-dimensional condition space (pH, temperature, spectral band) so that each condition held for many cells while *no* cell satisfied all three at once—an empty physical support. A per-cell detector returned nothing and a bulk-average detector could not report the truth, yet the categorical address (the intersection of the three condition regions) was non-empty and bounded, and conjunction individuation reconstructed the node’s region (Theorem 9.4, Theorem 9.6). The

ordered-sequence variant behaved identically. A control with non-empty support confirmed that per-cell detection succeeds precisely when the support is non-empty. Figure 5 displays the result.

14.6 Swarm coordination

Simulating the Kuramoto model with Lorentzian natural frequencies, the order parameter stayed near zero below $K_c = 2\sigma_\omega/\pi$ and rose toward $\sqrt{1 - K_c/K}$ above it, matching the mean-field branch at strong coupling (Theorem 8.3). In the phase-locked regime the residual phase spread fell below the cell width, so all agents shared one cell with zero coordination messages (Theorem 8.4); the distinguishable joint-state count $T(m, D) = D(D + 1)^{m-1}$ grew super-exponentially with channel count (Theorem 8.6); and dropping the coupling below K_c relaxed the coherence downward, the decoherence fault signal (Theorem 8.8). Figure 6 displays the result.

14.7 Closure and coherence

The composite catalytic power $1 - \prod_i (1 - \kappa_i)$ matched sequential residual reduction to machine precision and stayed strictly below one (Theorem 10.3). A support structure was robust to single-catalyst removal exactly when it contained a mutually-supporting group of at least three members—never at one or two (Theorem 10.4). A two-cluster configuration met a confidence threshold while remaining un-closed (Theorem 10.8), and every query terminated in convergent closure or honest decline (Theorem 10.10). Figure 7 displays the result.

Remark 14.1 (Status of the numerical witness). The figures confirm the results on concrete instances and display their quantitative form; they are evidence of correctness and magnitude, not a substitute for the proofs, which hold for every instance satisfying the premises. Where a result is asymptotic—the mean-field synchronisation branch of Theorem 8.3—the suite asserts what holds at finite size (incoherence below K_c , coherent onset above it, convergence to the prediction at strong coupling) and records the expected near-critical deviation rather than concealing it. Where the reference data are real and noisy—enzyme efficiencies against catalytic distance—

the suite tests the theorem’s actual claim, the inverse trend, not a strict monotonicity the scatter would not support. All thirty-four checks passed.

15 Discussion, Scope, and Conclusion

15.1 Relation to existing monitoring practice

The framework does not discard established bioprocess monitoring; it supplies the foundation that established practice lacks and recovers its components as special cases. Flux balance analysis (Orth et al., 2010) is the mass-balance operator $\mathcal{T}_{\text{bal}}$ of Theorem 6.3 without the thermodynamic and kinetic constraints; the framework adds cycle consistency and backward-address feasibility and, crucially, an existence-and- uniqueness proof for the reconstructed state (Theorem 6.6). Soft sensors (Luttmann et al., 2012; Mandenius and Gustavsson, 2015) are instances of trajectory completion, but with the empirically fitted error bar replaced by a theory-set uncertainty region bounded below by the resolution floor (Section 3). Multivariate statistical monitoring (Qin, 2012; Wold et al., 1987) detects departures in a learned subspace; the framework detects them as address escape in the network’s own coordinates, with a mechanistic rather than a correlational reading, and adds the swarm-coherence and contested-closure signals that have no statistical-monitoring analogue. Classical estimation (Kalman, 1960) propagates a point estimate with a covariance; the framework propagates a bounded region and reports, honestly, when the region cannot be reduced to one class. The Process Analytical Technology programme (U.S. Food and Drug Administration, 2004) calls for exactly the uncertainty-explicit, mechanism-grounded monitoring the framework provides.

15.2 Why the framework is not phenomenology

A reasonable concern is whether the framework merely re-labels known results. It does not: from the single premise of a finite observer against a non-completable medium, it *derives*

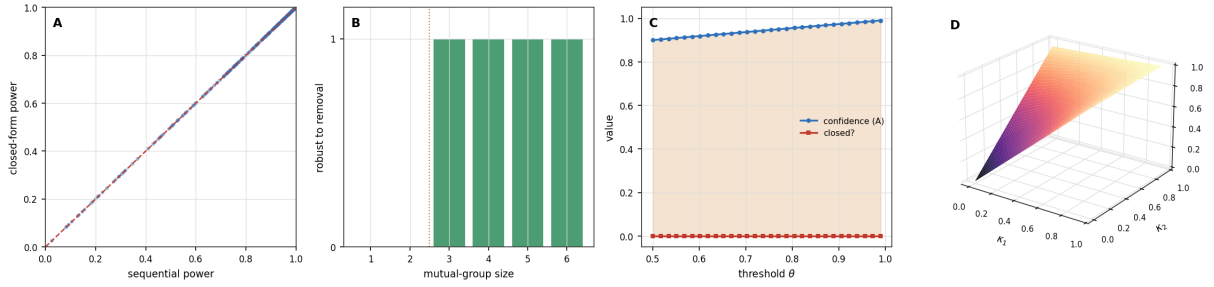


Figure 7: Catalytic Power, the Coherence Triangle, and Closure Beyond Confidence.

(A) Composite catalytic power computed by the closed form $1 - \prod_i (1 - \kappa_i)$ on the linear y -axis versus sequential residual reduction on the linear x -axis (0 to 1), over 400 random catalyst chains (steel-blue), with the identity $y = x$ marked (coral dashed). Every point lies on the diagonal (maximum absolute error 0.0): the multiplicative composition law of Theorem 10.3 is exact, and the composite power stays strictly below one. (B) Robustness to single-catalyst removal (0 or 1) versus mutual-support group size on the linear x -axis (1 to 6), bars with the $2 \rightarrow 3$ boundary marked (orange dotted); groups of size one and two are never robust (coral), groups of three and above always are (green). Robust grounding first appears at three mutually reinforcing catalysts and never below, the exact content of Theorem 10.4: a lone or paired source cannot survive the failure of one member. (C) Confidence value and closure status on the linear y -axis (0 to 1) versus threshold θ on the linear x -axis (0.5 to 0.99) for two-cluster configurations: cluster-A confidence (blue) stays at ~ 0.95 , above every threshold, while the closure flag (coral) stays at zero, with the gap shaded. A configuration can meet any confidence threshold yet remain not closed—an un-invoked catalyst reaches a distinct region—the strict-separation result of Theorem 10.8. (D) Three-dimensional surface of the two-catalyst composite power $1 - (1 - \kappa_1)(1 - \kappa_2)$ over the unit square $\kappa_1, \kappa_2 \in [0, 1]$, magma colourmap, viewing angle elevation 22° azimuth -55° . The surface rises from zero at the origin to one at (1, 1), saturating fastest along either axis and slowest on the diagonal where both catalysts are weak—the geometry of diminishing returns and the value of diverse, independent evidence (Theorem 10.3).

the resolution floor (Theorem 3.3), the convergence of state reconstruction (Theorem 6.6), the localisation and resolution laws (Theorems 7.2 and 7.3), the coordination and coherence results (Theorems 8.4 and 8.8), the closure criterion (Theorem 10.10), and the monitor–control separation (Theorems 11.1 to 11.3). Each is a consequence of the premise, not an assumption added to fit a case. Where the framework recovers classical results—Arrhenius and Eyring kinetics (Theorem 5.3), Kirchhoff-type conservation laws (Theorem 5.10)—it does so as corollaries, which is evidence for the premise rather than a restatement of the corollaries.

15.3 Scope and limitations

The framework is a monitoring theory and a monitoring algorithm; it is not a control law, a completed software system, or an experimental validation on a physical vessel, and the separation of monitoring from control (Section 11) is deliberate. Several assumptions bound its present reach. The trajectory-completion convergence (Theorem 6.6) assumes an autonomous, time-homogeneous network; strongly time-varying kinetics (rapid induction, phase transitions in the culture) require a non-autonomous extension that would forgo the time-invariance of backward addresses (Theorem 6.4). The localisation theorems (Section 7) assume a homogeneous propagation medium and a single event at a time; spatial heterogeneity and simultaneous events require, respectively, spatially varying propagation parameters and source separation. The swarm results (Section 8) assume controllable coupling and an all-to-all topology in the thermodynamic limit; finite-size and heterogeneous-topology corrections are needed for small arrays. The categorical modality’s contribution to localisation depends on maintaining distinguishable partition signatures over the relevant scale. None of these limitations touches the resolution floor or the monitor–control separation, which follow from the premise alone.

15.4 Conclusion

We have developed, from the single premise that a bioreactor is a population of finite observers against a non-completable medium, a

self-contained framework for bioprocess monitoring. The framework answers the three questions with which we began. A bioreactor measurement is the individuation of a bounded region of the culture’s state space, fixed by negation against the medium; it is never a point. The precision of monitoring is bounded below by a resolution floor that is a property of the finite culture and not of the instrument, so an honest monitor reports regions of width no less than that floor. And a monitoring query is entitled to report when it reaches closure—when no available further evidence can change its outcome—terminating either in a single identified region or in an explicit declaration that the evidence is contested.

Around these answers we built an operational pipeline: reconstruct the full metabolic state from partial observation by a provably convergent completion of the network’s own constraints; localise individual reactions by intersecting the arrival-time surfaces of several physically independent modalities, with resolution that multiplies as modalities are added and that is indifferent to optical opacity; coordinate a distributed monitoring swarm at zero overhead through phase-locking to the culture’s own oxygen-borne clock, reading the loss of coherence as an early fault; and terminate every query at closure, with contested closure exposing exactly where independent evidence diverges.

Finally, we established that monitoring must be kept separate from control, because a system that acts on the culture cannot be its unbiased observer, because the resolution floor forbids the point-precision a fused controller would presume, and because observation is closed under identification and does not extend to actuation. The dividend of the separation is the framework’s worst-case guarantee: with control absent or failed, the monitor still delivers detailed, uncertainty-bounded information about the process and never a wrong action. In the worst case, one simply has thorough knowledge of the bioreactor—which is precisely what a monitoring system should be built to guarantee.

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References

- Svante Arrhenius. Über die reaktionsgeschwindigkeit bei der inversion von rohrzucker durch säuren. *Zeitschrift für Physikalische Chemie*, 4:226–248, 1889.
- Peter Atkins, Julio de Paula, and James Keeler. *Atkins' Physical Chemistry*. Oxford University Press, 11th edition, 2018.
- Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922.
- Steven G. Boxer. Stark realities. *The Journal of Physical Chemistry B*, 113(10):2972–2983, 2009.
- Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- John Crank. *The Mathematics of Diffusion*. Oxford University Press, 2nd edition, 1979.
- Reinhard Diestel. *Graph Theory*. Springer, 5th edition, 2017.
- Henry Eyring. The activated complex in chemical reactions. *The Journal of Chemical Physics*, 3(2):107–115, 1935.
- Bertrand T. Fang. Simple solutions for hyperbolic and related position fixes. *IEEE Transactions on Aerospace and Electronic Systems*, 26(5):748–753, 1990.
- Lester R. Ford and Delbert R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956.
- Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.
- Reinhart Heinrich and Stefan Schuster. *The Regulation of Cellular Systems*. Chapman & Hall, 1996.
- Gerhard Herzberg. *Molecular Spectra and Molecular Structure. I. Spectra of Diatomic Molecules*. Van Nostrand, 2nd edition, 1950.
- Bertil Hille. *Ion Channels of Excitable Membranes*. Sinauer Associates, 3rd edition, 2001.
- Jacob N. Israelachvili. *Intermolecular and Surface Forces*. Academic Press, 3rd edition, 2011.
- Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957.
- Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.
- Edda Klipp, Wolfram Liebermeister, Christoph Wierling, and Axel Kowald. *Systems Biology: A Textbook*. Wiley-VCH, 2nd edition, 2016.
- Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.
- Nuno D. Lourenço, João A. Lopes, Cláudia F. Almeida, and Sarragu.
- Reiner Luttmann, Daniel G. Bracewell, Gesine Cornelissen, Krist V. Gernaey, Jarka Glassey, Volker C. Hass, Christoph Kaiser, Christiane Preusse, Gerald Striedner, and Carl-Fredrik Mandenius. Soft sensors in bioprocessing: A status report and recommendations. *Biotechnology Journal*, 7(8):1040–1048, 2012.
- Nancy A. Lynch. *Distributed Algorithms*. Morgan Kaufmann, 1996.
- Carl-Fredrik Mandenius and Robert Gustavsson. Mini-review: Soft sensors as means for PAT in the manufacture of bio-therapeutics. *Journal of Chemical Technology & Biotechnology*, 90(2):215–227, 2015.
- Leonor Michaelis and Maud L. Menten. Die kinetik der invertinwirkung. *Biochemische Zeitschrift*, 49:333–369, 1913.
- Jeffrey D. Orth, Ines Thiele, and Bernhard Ø. Palsson. What is flux balance analysis? *Nature Biotechnology*, 28(3):245–248, 2010.

- Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.
- S. Joe Qin. Survey on data-driven industrial process monitoring and diagnosis. *Annual Reviews in Control*, 36(2):220–234, 2012.
- Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- Bernhard Sonnleitner. Automated measurement and monitoring of bioprocesses: key elements of the M³C strategy. *Advances in Biochemical Engineering/Biotechnology*, 132:1–33, 2013.
- Steven H. Strogatz. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4):1–20, 2000.
- U.S. Food and Drug Administration. Guidance for industry: PAT—a framework for innovative pharmaceutical development, manufacturing, and quality assurance. *FDA Guidance Document*, 2004. Center for Drug Evaluation and Research.
- Nicolaas G. van Kampen. *Stochastic Processes in Physics and Chemistry*. North-Holland, 3rd edition, 2007.
- Venkat Venkatasubramanian, Raghunathan Rengaswamy, Kewen Yin, and Surya N. Kavuri. A review of process fault detection and diagnosis. Part I: Quantitative model-based methods. *Computers & Chemical Engineering*, 27(3):293–311, 2003.
- Andrew J. Viterbi. Error bounds for convolutional codes and an asymptotically optimum decoding algorithm. *IEEE Transactions on Information Theory*, 13(2):260–269, 1967.
- Marian von Smoluchowski. Versuch einer mathematischen theorie der koagulationskinetik kolloider lösungen. *Zeitschrift für Physikalische Chemie*, 92:129–168, 1917.
- Svante Wold, Kim Esbensen, and Paul Geladi. Principal component analysis. *Chemometrics and Intelligent Laboratory Systems*, 2(1–3):37–52, 1987.

Algorithm 1 Bioreactor Monitoring Cycle

Require: Reaction network $G = (V, E, w)$ with standard potentials; partial sensor readings $\mathcal{O}$; nominal reference addresses $\{\mathcal{A}_i^H\}$; catalyst registry; swarm phase stream $\{\phi_j\}$; floor β .

Ensure: State report $\tilde{\mathbf{X}}^*$ (bounded region per species); localised events; coherence R and regime; per-query verdict (identified region or DECLINE).

- 1: **Coordinates.** Map each sensor reading to S-entropy coordinates $\mathbf{s} = (s_k, s_t, s_e)$ (Theorem 4.1); individuate each against the medium by negation (Theorem 2.6).
 - 2: **Completion.** Initialise fuzzy states: measured nodes as narrow regions of width $\geq \beta$, unmeasured nodes as maximum-entropy priors over physiological ranges.
 - 3: **repeat**
 - 4: $\tilde{\mathbf{X}} \leftarrow \mathcal{T}_{\text{bal}}(\tilde{\mathbf{X}})$ $\triangleright$ mass balance at every node
 - 5: $\tilde{\mathbf{X}} \leftarrow \mathcal{T}_{\text{therm}}(\tilde{\mathbf{X}})$ $\triangleright$ thermodynamic cycle consistency
 - 6: $\tilde{\mathbf{X}} \leftarrow \mathcal{T}_{\text{kin}}(\tilde{\mathbf{X}})$ $\triangleright$ Viterbi backward-address feasibility
 - 7: **until** $d(\tilde{\mathbf{X}}, \tilde{\mathbf{X}}_{\text{prev}}) < \varepsilon$ $\triangleright$ contraction, Theorem 6.6
 - 8: **Health.** For each node, compare its backward address to the nominal $\mathcal{A}_i^H$; flag *address escape* where the categorical distance exceeds the per-node threshold (Theorem 6.5).
 - 9: **Localisation.** For each event of interest, intersect the arrival-time surfaces of the available modalities to recover $(\mathbf{r}_0, t_0)$ (Theorem 7.2); resolution multiplies with modality count (Theorem 7.3).
 - 10: **Coherence.** Estimate the swarm order parameter R from $\{\phi_j\}$; classify the coordination regime; if R has fallen a regime, raise the decoherence fault (Theorem 8.9).
 - 11: **Closure.** For each monitoring query, invoke catalysts until no available catalyst adds a new state-region (Theorem 10.7); require a ≥ 3 -catalyst coherence cycle for a grounded verdict (Theorem 10.4).
 - 12: **if** $\mathcal{C}$ is a single class **then**
 - 13: **return** identified region (convergent closure)
 - 14: **else**
 - 15: **return** DECLINE with the distinct regions (contested closure, Theorem 10.10)
 - 16: **end if**
 - 17: **(No actuation.)** Hand the report across the interface to the separate control framework (Section 11).
-